Survival Regression Analysis

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Learning Objectives

- Visualize cumulative incidence of cancer deaths and other deaths in cancer patients.
- Use Cox proportional hazards models to compare mortality rates among different demographic groups.
- Calculate, visualize and interpret time-varying hazard ratios.

1. Import and manage data

```
data <- read.csv("melanoma_survival.csv", header=TRUE)

# Age
table(data$Age.recode.with..1.year.olds.and.90.)
```

```
15-19 years 20-24 years 25-29 years 30-34 years 35-39 years 40-44 years
       1412        3908        7302       11183       14862       19325
45-49 years 50-54 years 55-59 years 60-64 years 65-69 years 70-74 years
      26097       33805       41309       47299       50594       47770
75-79 years 80-84 years 85-89 years   90+ years
      42301       34539       22827       12385
```

```
data$age <- dplyr::case_when(
  data[[1]] %in% c("15-19 years","20-24 years","25-29 years","30-34 years",
                   "35-39 years","40-44 years","45-49 years") ~ "<50",
  data[[1]] %in% c("50-54 years","55-59 years") ~ "50-59",
  data[[1]] %in% c("60-64 years","65-69 years") ~ "60-69",
  data[[1]] %in% c("70-74 years","75-79 years","80-84 years","85-89 years","90+ years") ~ "70+",
  TRUE ~ NA_character_
)
data$age <- factor(data$age, levels = c("70+","<50","50-59","60-69"))

# Race
colnames(data)[2] <- "race"
data$race <- factor(data$race)
data$race <- relevel(data$race, ref = "Non-Hispanic White")

# Sex
colnames(data)[3] <- "sex"
data$sex <- factor(data$sex, levels = c("Female","Male"))
```

```
# Create Head_and_Neck classification and update subsite factor levels
# Rename column 5 to "subsite"
colnames(data)[5] = "subsite"

# Recode subsite into broader categories
data <- data %>%
  mutate(
    subsite = case_when(
      subsite %in% c("C44.1-Eyelid", "C44.2-External ear",
                     "C44.3-Skin other/unspec parts of face",
                     "C44.4-Skin of scalp and neck") ~ "Head and Neck",
      subsite %in% c("C44.5-Trunk") ~ "Trunk",
      subsite %in% c("C44.6-Upper limb") ~ "Arm",
      subsite %in% c("C44.7-Lower limb") ~ "Leg",
      TRUE ~ subsite
    ),
    subsite = factor(subsite, levels = c("Head and Neck", "Trunk", "Arm", "Leg"))
)

# Year of diagnosis
data$yod <- ifelse(data[[4]] < 2010, "2004-2009", "2010-2022")
data$yod <- factor(data$yod, levels = c("2004-2009","2010-2022"))

# Stage
colnames(data)[6]="stage"
table(data$stage)
```

```
                          Distant site(s)/node(s) involved
                                                      8504
                                                   In situ
                                                        47
                                           Localized only
                                                    345988
Regional by both direct extension and lymph node involvement
                                                      4340
                           Regional by direct extension only
                                                     11179
                          Regional lymph nodes involved only
                                                     19852
                           Unknown/unstaged/unspecified/DCO
                                                     27008
```

```
data$stage <- dplyr::case_when(
  data$stage %in% c("Blank(s)", "In situ", "Unknown/unstaged/unspecified/DCO") ~ "Unstaged",
  data$stage %in% c("Regional by both direct extension and lymph node involvement",
                    "Regional by direct extension only",
                    "Regional lymph nodes involved only") ~ "Regional",
  data$stage == "Distant site(s)/node(s) involved" ~ "Distant",
  data$stage == "Localized only" ~ "Localized",
```

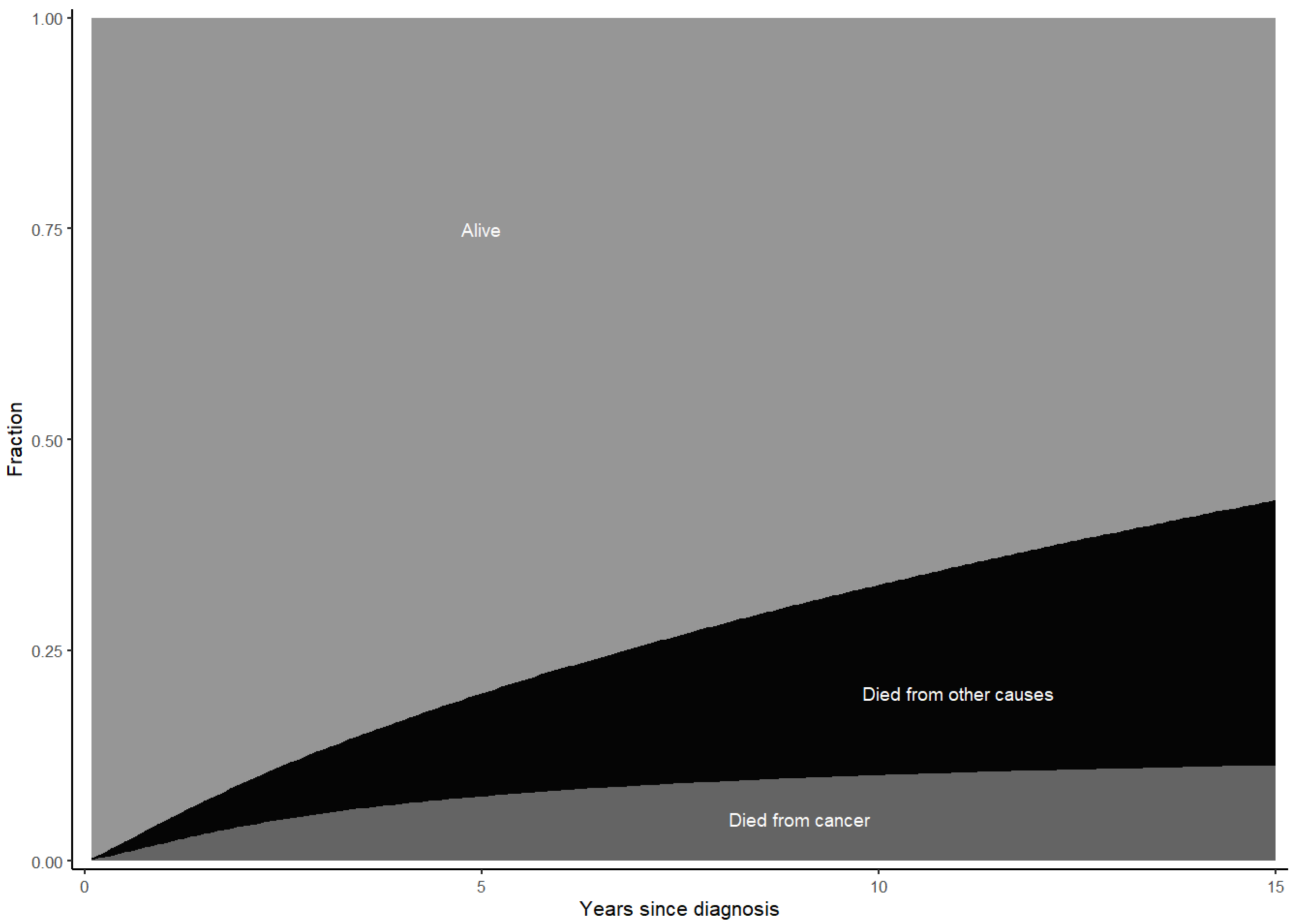
```
  TRUE ~ data$stage
)
data$stage <- factor(data$stage, levels = c("Localized","Regional","Distant","Unstaged"))

# Survival time
colnames(data)[9] <- "time"
data$time <- as.numeric(data$time)
data$time <- data$time / 12  # convert months to years

# Survival event
data$event <- dplyr::case_when(
  data[[7]] == "Dead (attributable to this cancer dx)" ~ "Cancer death",
  data[[8]] == "Dead (attributable to causes other than this cancer dx)" ~ "Other death",
  data[[7]] == "Alive or dead of other cause" & data[[8]] == "Alive or dead due to cancer" ~ "Ali
  TRUE ~ NA_character_
)
```

2. Plot cumulative incidence using the survival package

```
# Create a multistate (competing risks) survival object with no covariates (~1)
sf_HN_MS <- survfit(Surv(time=data$time,event=data$event,type="mstate")~1)
DATA <- matrix(NA,nrow=3*length(sf_HN_MS$time),ncol=3)

# Time for each of "Cancer death", "other death", and "alive"
DATA[,1] <- rep(sf_HN_MS$time,3)

# Probabilities are cumulative. These are where "Cancer death", "other death", and "alive" each s
DATA[,2] <- c(rep(0,length(sf_HN_MS$time)),sf_HN_MS$pstate[,2],sf_HN_MS$pstate[,2]+sf_HN_MS$pstat

# Probabilities are cumulative. These are where "Cancer death", "other death", and "alive" each e
DATA[,3] <- c(sf_HN_MS$pstate[,2],sf_HN_MS$pstate[,2]+sf_HN_MS$pstate[,3],rep(1,length(sf_HN_MS$t
DATA <- as.data.frame(DATA)

colnames(DATA) <- c("x","ymin","ymax")
DATA$state <- c(rep("Cancer Death",length(sf_HN_MS$time)),rep("Other death",length(sf_HN_MS$time)
             rep("Alive",length(sf_HN_MS$time)))
DATA$state <- factor(DATA$state,levels=c("Cancer Death","Other death","Alive"))

p0 <- ggplot(data = DATA, aes(x = x, y = ymin, ymin = ymin, ymax = ymax, fill = state)) +
  scale_fill_manual(values=c("gray40", "gray2", "grey60"), guide_legend(title = "Age at diagnosis
  geom_ribbon(aes(fill = state,linetype = NA), alpha = 1, show.legend = FALSE) +
  theme_bw(base_size = 8) + labs(x = 'Years since diagnosis', y = 'Fraction') +
  theme(axis.title.y = element_text(vjust=1.5))+
  theme(legend.position.inside = c(0.3,0.8),legend.key = element_blank(),
        legend.background = element_blank(), legend.text = element_text(size=7),
        legend.key.height = unit(0.05,"inch")) +
  theme(panel.border = element_blank(),
        panel.background = element_blank(),
```

```
        panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(),
        axis.line.x = element_line(color = "black", linewidth = 0.4),
        axis.line.y = element_line(color = "black", linewidth = 0.4))+
  annotate("text", x = c(5, 9, 11), y = c(0.75, 0.05, 0.2), label = c("Alive", "Died from cancer",
                                                                     "Died from other causes"),
                                                              color = "White", size = 2.5) +
  scale_x_continuous(limits = c(0, 15), expand = (c(0.01, 0.01)), breaks = c(0, 5, 10, 15, 20)) +
  scale_y_continuous(limits = c(0, 1), expand = c(0.005,0.005))

p0
```

```
Warning: Removed 141 rows containing missing values or values outside the scale range
(`geom_ribbon()`).
```



3. Plotting cumulative incidence by covariate

Here, we create a `survift` object with cumulative incidence stratified by the levels of our variables of interest. In order to extract information for plotting from the `survfit` objects, we need to know the order of the levels and how many rows are associated with each level.

Let's start by looking at sex and race & ethnicity.

```
data$time <- as.numeric(as.character(data$time))

# Cumulative incidence of cancer death by sex
sf_sex <- survfit(Surv(time=data$time,event=data$event,type="mstate")~data$sex)

# Cumulative incidence of cancer death by race and ethnicity
sf_race_eth <- survfit(Surv(time=data$time,event=data$event,type="mstate")~data$race)

# Cumulative incidence of cancer death by subsite
sf_subsite <- survfit(Surv(time=data$time,event=data$event,type="mstate")~data$subsite)

# Cumulative incidence of cancer death by age
sf_age <- survfit(Surv(time=data$time,event=data$event,type="mstate")~data$age)

# Check the strata of each model
sf_sex$strata
```

```
data$sex=Female   data$sex=Male
            227             227
```

```
sf_race_eth$strata
```

```
                        data$race=Non-Hispanic White
                                                 227
                       data$race=Hispanic (All Races)
                                                 227
data$race=Non-Hispanic American Indian/Alaska Native
                                                 213
    data$race=Non-Hispanic Asian or Pacific Islander
                                                 227
                        data$race=Non-Hispanic Black
                                                 227
                 data$race=Non-Hispanic Unknown Race
                                                 227
```

```
sf_subsite$strata
```

```
data$subsite=Head and Neck         data$subsite=Trunk
                       227                        227
          data$subsite=Arm           data$subsite=Leg
                       227                        227
```

```
sf_age$strata
```

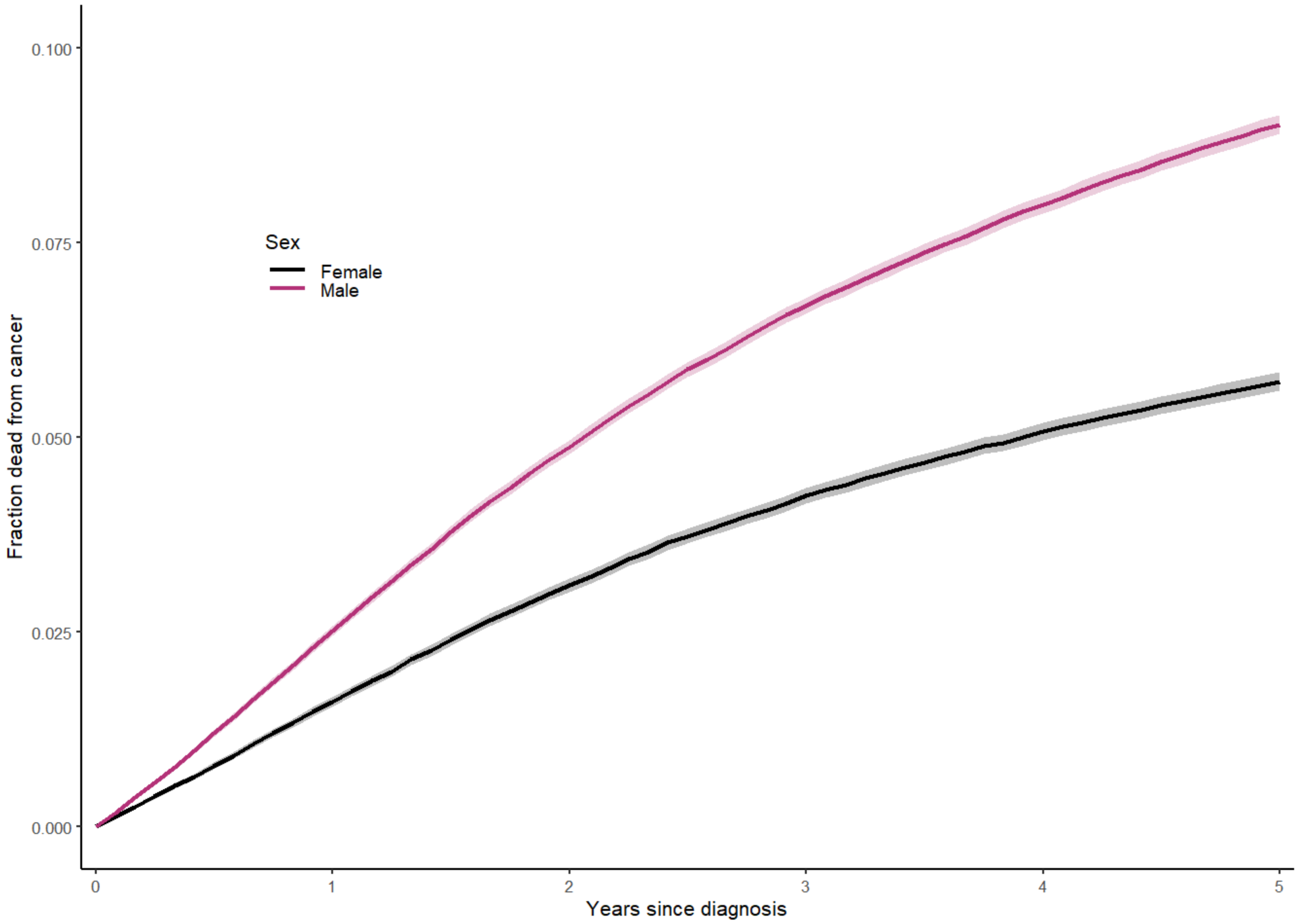
```
  data$age=70+   data$age=<50 data$age=50-59 data$age=60-69
           227            227            227            227
```

In the example data, most groups have information across all possible survival months, but that isn't true for small subgroups (in the example data, Non-Hispanic American Indian/Alaskan Native).

```
# Plot cumulative incidence of cancer death by sex
# For ease of plotting for other covariates, use generic plot code after setting
df_sex <- sf_sex
mycolors <- magma

# Create a data matrix with the correct number of rows:
# A 0 for each group plus the sum of the number of rows for each subgroup
DATA_ci_sex <- matrix(NA,nrow=(length(df_sex$strata)+sum(df_sex$strata)),ncol=4)
# x
DATA_ci_sex[,1] <- c(0,df_sex$time[1:df_sex$strata[1]],
           0,df_sex$time[(1+df_sex$strata[1]):(sum(df_sex$strata[1:2]))])
# y
DATA_ci_sex[,2] <- c(0,df_sex$pstate[1:df_sex$strata[1],2],
           0,df_sex$pstate[(1+df_sex$strata[1]):(sum(df_sex$strata[1:2])),2])
# 95% CI (lower bound)
DATA_ci_sex[,3] <- c(0,df_sex$lower[1:df_sex$strata[1],2],
           0,df_sex$lower[(1+df_sex$strata[1]):(sum(df_sex$strata[1:2])),2])
# 95% CI (upper bound)
DATA_ci_sex[,4] <- c(0,df_sex$upper[1:df_sex$strata[1],2],
           0,df_sex$upper[(1+df_sex$strata[1]):(sum(df_sex$strata[1:2])),2])
DATA_ci_sex <- as.data.frame(DATA_ci_sex)
colnames(DATA_ci_sex) <- c("x","y","ymin","ymax")
breaks <- which(c(DATA_ci_sex$x,0)==0)[-1]
DATA_ci_sex$sex <- c(rep("Female",df_sex$strata[1]+1),rep("Male",df_sex$strata[2]+1))
DATA_ci_sex$sex <- factor(DATA_ci_sex$sex,levels=levels(data$sex))

# In this plot, we have two groups, so it gets busy to also show other
# cause deaths. We drop them from this plot.
plot_sex <- ggplot(data=DATA_ci_sex,aes(x = x, y = y, ymin = ymin, ymax = ymax, color = sex)) +
  scale_color_manual(values=mycolors(3)[-3],name ="Sex") +
  geom_ribbon(aes(fill = sex, linetype = NA), alpha = 0.25, show.legend=FALSE) +
  geom_line(linewidth = 0.75) +
  scale_fill_manual(guide = "none", values = mycolors(3)[-3]) +
  theme_bw(base_size = 8) + labs(x = 'Years since diagnosis', y = 'Fraction dead from cancer') +
  theme(axis.title.y = element_text(vjust=1.5)) +
  theme(legend.position = "inside",
        legend.position.inside = c(0.2,0.7), legend.key = element_blank(),
        legend.background = element_blank(), legend.text = element_text(size = 7),
        legend.key.height = unit(0.05,"inch")) +
  theme(panel.border = element_blank(),
        panel.background = element_blank(),
        panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(),
        axis.line.x = element_line(color = "black", linewidth = 0.4),
        axis.line.y = element_line(color = "black", linewidth = 0.4)) +
  scale_x_continuous(limits = c(0, 5), expand = (c(0.01, 0.01))) +
  scale_y_continuous(limits=c(0, 0.1), expand =c(0.005, 0.005))
```

```
plot_sex
```

```
Warning: Removed 334 rows containing missing values or values outside the scale range
(`geom_ribbon()`).

Warning: Removed 334 rows containing missing values or values outside the scale range
(`geom_line()`).
```

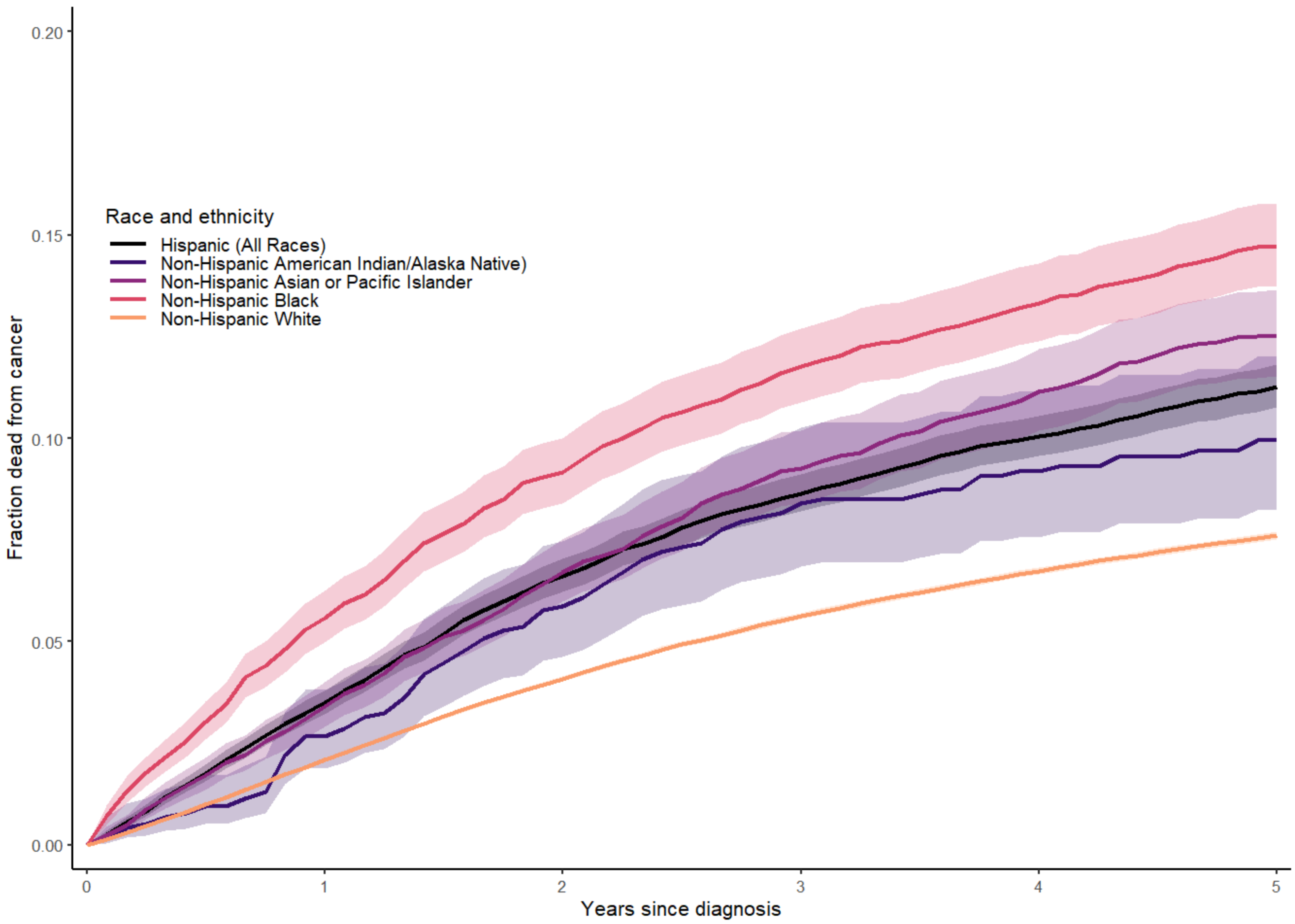



```
# Plot cumulative incidence of cancer death by race and ethnicity
# For ease of plotting for other covariates, use generic plot code after setting
df_re <- sf_race_eth
# Create a data matrix with the correct number of rows:
# A 0 for each group plus the sum of the number of rows for each subgroup.
# Drop "unknown" race.
DATA_ci_re=matrix(NA,nrow=(5+sum(df_re$strata[1:5])),ncol=4)
# x
DATA_ci_re[,1]=c(0,df_re$time[1:df_re$strata[1]],
           0,df_re$time[(1+df_re$strata[1]):(sum(df_re$strata[1:2]))],
           0,df_re$time[(1+sum(df_re$strata[1:2])):(sum(df_re$strata[1:3]))],
           0,df_re$time[(1+sum(df_re$strata[1:3])):(sum(df_re$strata[1:4]))],
           0,df_re$time[(1+sum(df_re$strata[1:4])):(sum(df_re$strata[1:5]))])
# y
```

```
DATA_ci_re[,2]=c(0,df_re$pstate[1:df_re$strata[1],2],
           0,df_re$pstate[(1+df_re$strata[1]):(sum(df_re$strata[1:2])),2],
           0,df_re$pstate[(1+sum(df_re$strata[1:2])):(sum(df_re$strata[1:3])),2],
           0,df_re$pstate[(1+sum(df_re$strata[1:3])):(sum(df_re$strata[1:4])),2],
           0,df_re$pstate[(1+sum(df_re$strata[1:4])):(sum(df_re$strata[1:5])),2])
# 95% CI (lower bound)
DATA_ci_re[,3]=c(0,df_re$lower[1:df_re$strata[1],2],
           0,df_re$lower[(1+df_re$strata[1]):(sum(df_re$strata[1:2])),2],
           0,df_re$lower[(1+sum(df_re$strata[1:2])):(sum(df_re$strata[1:3])),2],
           0,df_re$lower[(1+sum(df_re$strata[1:3])):(sum(df_re$strata[1:4])),2],
           0,df_re$lower[(1+sum(df_re$strata[1:4])):(sum(df_re$strata[1:5])),2])
# 95% CI (upper bound)
DATA_ci_re[,4]=c(0,df_re$upper[1:df_re$strata[1],2],
           0,df_re$upper[(1+df_re$strata[1]):(sum(df_re$strata[1:2])),2],
           0,df_re$upper[(1+sum(df_re$strata[1:2])):(sum(df_re$strata[1:3])),2],
           0,df_re$upper[(1+sum(df_re$strata[1:3])):(sum(df_re$strata[1:4])),2],
           0,df_re$upper[(1+sum(df_re$strata[1:4])):(sum(df_re$strata[1:5])),2])

DATA_ci_re=as.data.frame(DATA_ci_re)
colnames(DATA_ci_re)=c("x","y","ymin","ymax")
breaks=which(c(DATA_ci_re$x,0)==0)[-1]
DATA_ci_re$race_eth=c(rep("Non-Hispanic White",   1+df_re$strata[1]),
           rep("Hispanic (All Races)", 1+ df_re$strata[2]),
           rep("Non-Hispanic American Indian/Alaska Native)", 1+ df_re$strata[3]),
           rep("Non-Hispanic Asian or Pacific Islander", 1+ df_re$strata[4]),
           rep("Non-Hispanic Black", 1+ df_re$strata[5]))
DATA_ci_re$race_eth=factor(DATA_ci_re$race_eth)

# In this plot, we have two groups, so it gets busy to also show other
# cause deaths. We drop them from this plot.
plot_race_eth = ggplot(data=DATA_ci_re,aes(x=x,y=y,ymin=ymin,ymax=ymax,color=race_eth))+
  scale_color_manual(values=mycolors(6)[-6],name ="Race and ethnicity")+
  geom_ribbon(aes(fill=race_eth,linetype=NA),alpha=0.25,show.legend=FALSE)+
  geom_line(linewidth=0.75)+
  scale_fill_manual(guide="none",values=mycolors(6)[-6])+
  theme_bw(base_size = 8) + labs(x='Years since diagnosis', y='Fraction dead from cancer')+
  theme(axis.title.y=element_text(vjust=1.5))+
  theme(legend.position = "inside",
        legend.position.inside = c(0.2,0.7),legend.key = element_blank(),
        legend.background=element_blank(),legend.text=element_text(size=7),
        legend.key.height=unit(0.05,"inch"))+
  theme(panel.border = element_blank(),
        panel.background = element_blank(),
        panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(),
        axis.line.x = element_line(color = "black",linewidth=0.4),
        axis.line.y = element_line(color = "black",linewidth=0.4))+
  scale_x_continuous(limits=c(0,5),expand=(c(0.01,0.01)))+
  scale_y_continuous(limits=c(0,.2),expand=c(0.005,0.005))
```

```
plot_race_eth
```

```
Warning: Removed 821 rows containing missing values or values outside the scale range
(`geom_ribbon()`).

Warning: Removed 821 rows containing missing values or values outside the scale range
(`geom_line()`).
```

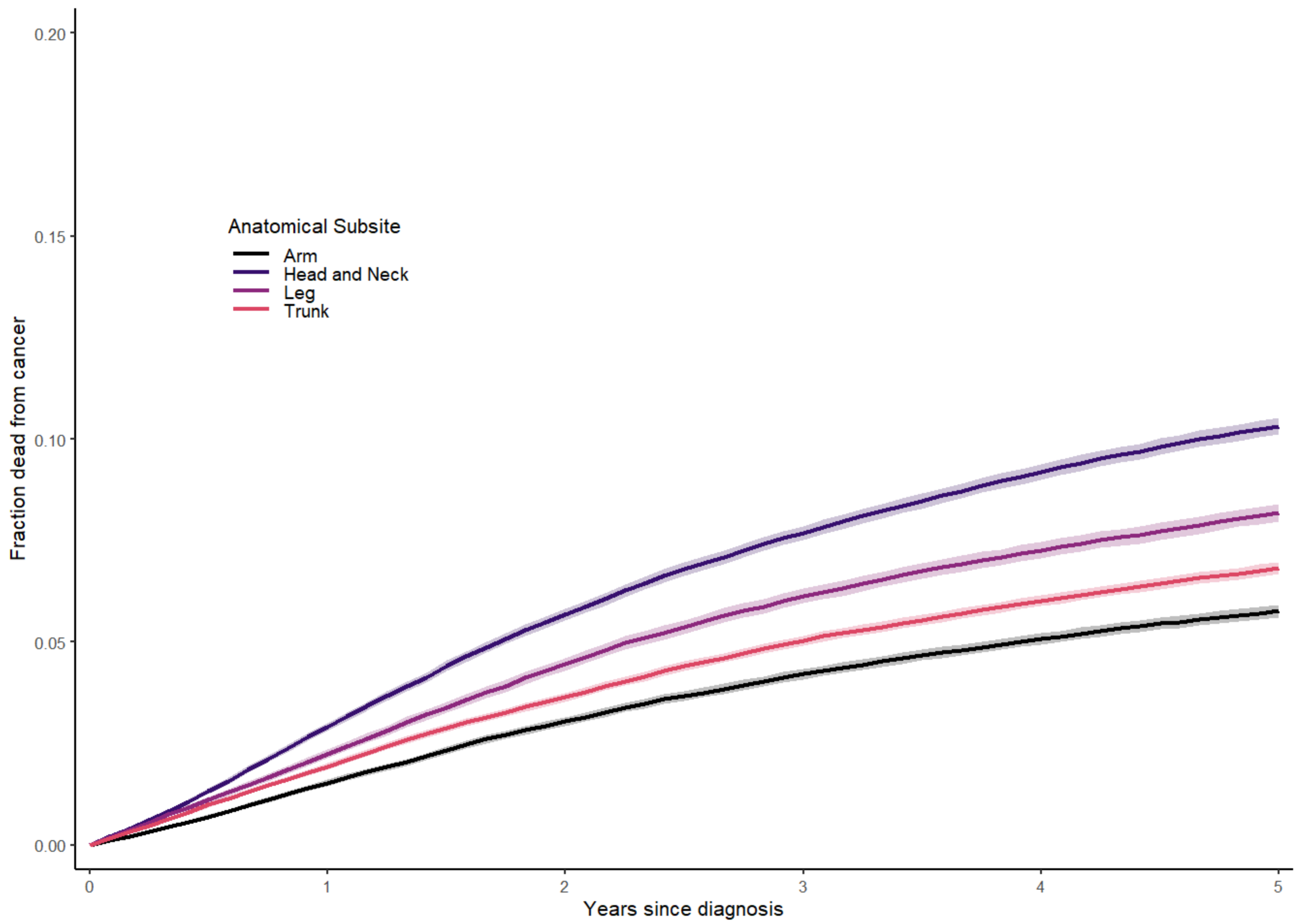



```
# Plot cumulative incidence of cancer death by subsite
# For ease of plotting for other covariates, use generic plot code after setting
df_subsite <- sf_subsite
# Create a data matrix with the correct number of rows:
# A 0 for each group plus the sum of the number of rows for each subgroup.
# Drop "unknown" race.
DATA_ci_subsite=matrix(NA,nrow=(4+sum(df_subsite$strata[1:4])),ncol=4)
# x
DATA_ci_subsite[,1] <- c(0,df_subsite$time[1:df_subsite$strata[1]],
           0,df_subsite$time[(1+df_subsite$strata[1]):(sum(df_subsite$strata[1:2]))],
           0,df_subsite$time[(1+sum(df_subsite$strata[1:2])):(sum(df_subsite$strata[1:3]))],
           0,df_subsite$time[(1+sum(df_subsite$strata[1:3])):(sum(df_subsite$strata[1:4]))])
# y
DATA_ci_subsite[,2] <- c(0,df_subsite$pstate[1:df_subsite$strata[1],2],
```

```
          0,df_subsite$pstate[(1+df_subsite$strata[1]):(sum(df_subsite$strata[1:2])),2],
          0,df_subsite$pstate[(1+sum(df_subsite$strata[1:2])):(sum(df_subsite$strata[1:3])),2],
          0,df_subsite$pstate[(1+sum(df_subsite$strata[1:3])):(sum(df_subsite$strata[1:4])),2])
#95% CI (lower bound)
DATA_ci_subsite[,3] <- c(0,df_subsite$lower[1:df_subsite$strata[1],2],
          0,df_subsite$lower[(1+df_subsite$strata[1]):(sum(df_subsite$strata[1:2])),2],
          0,df_subsite$lower[(1+sum(df_subsite$strata[1:2])):(sum(df_subsite$strata[1:3])),2],
          0,df_subsite$lower[(1+sum(df_subsite$strata[1:3])):(sum(df_subsite$strata[1:4])),2])
#95% CI (upper bound)
DATA_ci_subsite[,4] <- c(0,df_subsite$upper[1:df_subsite$strata[1],2],
          0,df_subsite$upper[(1+df_subsite$strata[1]):(sum(df_subsite$strata[1:2])),2],
          0,df_subsite$upper[(1+sum(df_subsite$strata[1:2])):(sum(df_subsite$strata[1:3])),2],
          0,df_subsite$upper[(1+sum(df_subsite$strata[1:3])):(sum(df_subsite$strata[1:4])),2])
DATA_ci_subsite <- as.data.frame(DATA_ci_subsite)
colnames(DATA_ci_subsite) <- c("x","y","ymin","ymax")
breaks <- which(c(DATA_ci_subsite$x,0) == 0)[-1]
DATA_ci_subsite$subsite <- c(rep("Head and Neck", 1+df_subsite$strata[1]),
          rep("Trunk", 1+ df_subsite$strata[2]),
          rep("Arm", 1+ df_subsite$strata[3]),
          rep("Leg", 1+ df_subsite$strata[4]))
DATA_ci_subsite$subsite = factor(DATA_ci_subsite$subsite)

# In this plot, we have two groups, so it gets busy to also show other
# cause deaths. We drop them from this plot.
plot_subsite <- ggplot(data = DATA_ci_subsite, aes(x = x, y = y, ymin = ymin, ymax = ymax, color =
  scale_color_manual(values = mycolors(6)[-6],name ="Anatomical Subsite") +
  geom_ribbon(aes(fill = subsite, linetype = NA), alpha = 0.25, show.legend = FALSE) +
  geom_line(linewidth = 0.75) +
  scale_fill_manual(guide="none",values=mycolors(6)[-6]) +
  theme_bw(base_size = 8) + labs(x='Years since diagnosis', y='Fraction dead from cancer') +
  theme(axis.title.y=element_text(vjust=1.5)) +
  theme(legend.position = "inside",
        legend.position.inside = c(0.2,0.7), legend.key = element_blank(),
        legend.background = element_blank(), legend.text = element_text(size = 7),
        legend.key.height = unit(0.05,"inch")) +
  theme(panel.border = element_blank(),
        panel.background = element_blank(),
        panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(),
        axis.line.x = element_line(color = "black", linewidth = 0.4),
        axis.line.y = element_line(color = "black", linewidth = 0.4))+
  scale_x_continuous(limits = c(0,5), expand=(c(0.01, 0.01))) +
  scale_y_continuous(limits = c(0,0.2), expand=c(0.005, 0.005))

plot_subsite
```

```
Warning: Removed 668 rows containing missing values or values outside the scale range
(`geom_ribbon()`).
```

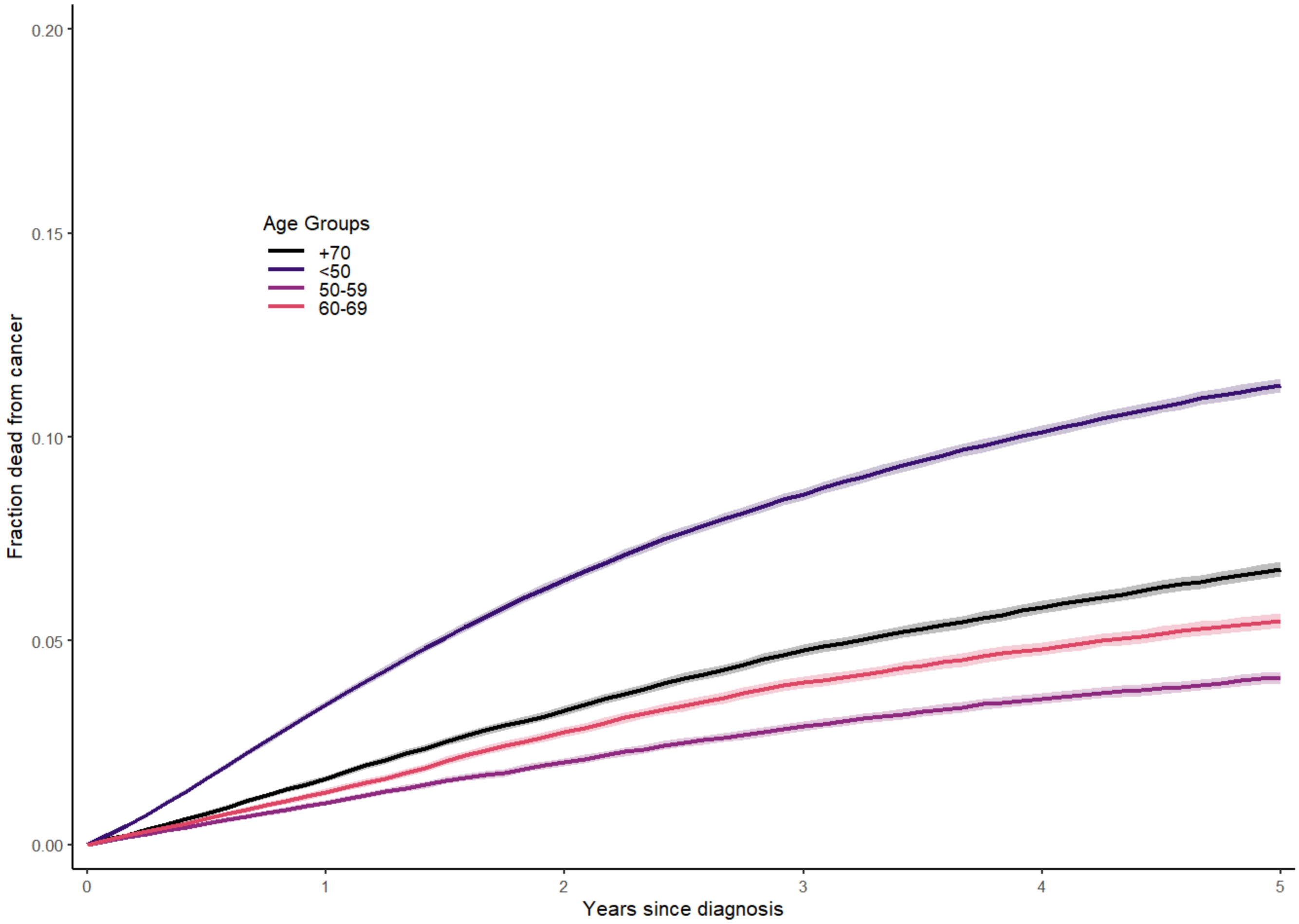
```
Warning: Removed 668 rows containing missing values or values outside the scale range
(`geom_line()`).
```




```
# Plot cumulative incidence of cancer death by race and ethnicity
# For ease of plotting for other covariates, use generic plot code after setting
df_age <- sf_age

# Create a data matrix with the correct number of rows:
# A 0 for each group plus the sum of the number of rows for each subgroup.
# Drop "unknown" race.
DATA_ci_age <- matrix(NA,nrow=(4+sum(df_age$strata[1:4])),ncol=4)
# x
DATA_ci_age[,1] <- c(0,df_age$time[1:df_age$strata[1]],
           0,df_age$time[(1+df_age$strata[1]):(sum(df_age$strata[1:2]))],
           0,df_age$time[(1+sum(df_age$strata[1:2])):(sum(df_age$strata[1:3]))],
           0,df_age$time[(1+sum(df_age$strata[1:3])):(sum(df_age$strata[1:4]))])
# y
DATA_ci_age[,2] <- c(0,df_age$pstate[1:df_age$strata[1],2],
           0,df_age$pstate[(1+df_age$strata[1]):(sum(df_age$strata[1:2])),2],
           0,df_age$pstate[(1+sum(df_age$strata[1:2])):(sum(df_age$strata[1:3])),2],
           0,df_age$pstate[(1+sum(df_age$strata[1:3])):(sum(df_age$strata[1:4])),2])
# 95% CI (lower bound)
DATA_ci_age[,3] <- c(0,df_age$lower[1:df_age$strata[1],2],
```

```
          0,df_age$lower[(1+df_age$strata[1]):(sum(df_age$strata[1:2])),2],
          0,df_age$lower[(1+sum(df_age$strata[1:2])):(sum(df_age$strata[1:3])),2],
          0,df_age$lower[(1+sum(df_age$strata[1:3])):(sum(df_age$strata[1:4])),2])
# 95% CI (upper bound)
DATA_ci_age[,4] <- c(0,df_age$upper[1:df_age$strata[1],2],
          0,df_age$upper[(1+df_age$strata[1]):(sum(df_age$strata[1:2])),2],
          0,df_age$upper[(1+sum(df_age$strata[1:2])):(sum(df_age$strata[1:3])),2],
          0,df_age$upper[(1+sum(df_age$strata[1:3])):(sum(df_age$strata[1:4])),2])
DATA_ci_age <- as.data.frame(DATA_ci_age)
colnames(DATA_ci_age) <- c("x","y","ymin","ymax")
breaks <- which(c(DATA_ci_age$x,0) == 0)[-1]
DATA_ci_age$age <- c(rep("<50", 1+df_age$strata[1]),
          rep("50-59", 1+ df_age$strata[2]),
          rep("60-69", 1+ df_age$strata[3]),
          rep("+70", 1+ df_age$strata[4]))
DATA_ci_age$age <- factor(DATA_ci_age$age)

# In this plot, we have two groups, so it gets busy to also show other
# cause deaths. We drop them from this plot.
plot_age <- ggplot(data = DATA_ci_age, aes(x = x, y = y, ymin = ymin, ymax = ymax, color = age)) +
  scale_color_manual(values = mycolors(6)[-6], name ="Age Groups") +
  geom_ribbon(aes(fill = age, linetype = NA), alpha = 0.25, show.legend = FALSE) +
  geom_line(linewidth = 0.75) +
  scale_fill_manual(guide="none", values = mycolors(6)[-6]) +
  theme_bw(base_size = 8) + labs(x = 'Years since diagnosis', y = 'Fraction dead from cancer') +
  theme(axis.title.y = element_text(vjust = 1.5)) +
  theme(legend.position = "inside",
        legend.position.inside = c(0.2,0.7),legend.key = element_blank(),
        legend.background = element_blank(), legend.text = element_text(size=7),
        legend.key.height=unit(0.05,"inch")) +
  theme(panel.border = element_blank(),
        panel.background = element_blank(),
        panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(),
        axis.line.x = element_line(color = "black", linewidth = 0.4),
        axis.line.y = element_line(color = "black", linewidth = 0.4))+
  scale_x_continuous(limits = c(0, 5), expand = (c(0.01, 0.01)))+
  scale_y_continuous(limits = c(0, 0.2), expand = c(0.005, 0.005))

plot_age
```

```
Warning: Removed 668 rows containing missing values or values outside the scale range
(`geom_ribbon()`).

Warning: Removed 668 rows containing missing values or values outside the scale range
(`geom_line()`).
```



4/ Cox proportional hazards models

We saw in the above plots that covariates impact the survival. If we assume that the rates are proportional over time, what are the hazard ratios?

We will focus on survival in the first 5 years.

```
# Define a censoring variable at 5 years survival
data$censored.time <- ifelse(data$time > 5, 5, data$time)
data$censored.event <- ifelse(data$time > 5, "Alive", as.character(data$event))

# Because we are considering the instantaneous rate of cancer death,
# we treat other deaths as being censored
data$cancer.death <- (data$censored.event == "Cancer death")

# Generate survival object
cancer.death.surv.5yr <- Surv(data$censored.time, data$cancer.death)

# By sex male
fit.coxph.sex <- coxph(cancer.death.surv.5yr ~ sex,
                       data = data)
summary(fit.coxph.sex)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ sex, data = data)

  n= 416918, number of events= 27689

           coef exp(coef) se(coef)     z Pr(>|z|)
sexMale 0.49774   1.64500  0.01306 38.12   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

        exp(coef) exp(-coef) lower .95 upper .95
sexMale     1.645     0.6079     1.603     1.688

Concordance= 0.557  (se = 0.001 )
Likelihood ratio test= 1538  on 1 df,   p=<2e-16
Wald test            = 1453  on 1 df,   p=<2e-16
Score (logrank) test = 1484  on 1 df,   p=<2e-16
```

```
# The hazard ratio is exp(ceof). Is it significant?

# Stage is the most important predictor of stage, so other variables are
# vulnerable to confounding. We should adjust for stage.
fit.coxph.sex_stage <- coxph(cancer.death.surv.5yr ~ sex + stage,
                             data = data)
summary(fit.coxph.sex_stage)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ sex + stage, data = data)

  n= 416918, number of events= 27689

                  coef exp(coef) se(coef)      z Pr(>|z|)
sexMale        0.40096   1.49325  0.01308  30.66   <2e-16 ***
stageRegional  2.07687   7.97947  0.01394 148.97   <2e-16 ***
stageDistant   3.20209  24.58386  0.01761 181.84   <2e-16 ***
stageUnstaged  0.69945   2.01265  0.02493  28.06   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

              exp(coef) exp(-coef) lower .95 upper .95
sexMale           1.493    0.66968     1.455     1.532
stageRegional     7.979    0.12532     7.764     8.201
stageDistant     24.584    0.04068    23.750    25.447
stageUnstaged     2.013    0.49686     1.917     2.113

Concordance= 0.748  (se = 0.002 )
Likelihood ratio test= 33986  on 4 df,   p=<2e-16
Wald test            = 44341  on 4 df,   p=<2e-16
Score (logrank) test = 78629  on 4 df,   p=<2e-16
```

```r
fit.coxph.sex_full <- coxph(cancer.death.surv.5yr ~ sex + stage + age + yod + race + event + subs:
                            data = data)
```

```
Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :
Loglik converged before variable 14 ; coefficient may be infinite.
```

```r
summary(fit.coxph.sex_full)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ sex + stage + age + yod +
    race + event + subsite, data = data)

  n= 415909, number of events= 27536
   (1009 observations deleted due to missingness)

                                               coef  exp(coef)   se(coef)
sexMale                                   1.835e-02  1.019e+00  1.349e-02
stageRegional                             5.586e-01  1.748e+00  1.420e-02
stageDistant                              1.358e+00  3.889e+00  1.806e-02
stageUnstaged                             3.828e-01  1.466e+00  2.509e-02
age<50                                   -4.221e-01  6.557e-01  2.042e-02
age50-59                                 -4.010e-01  6.696e-01  1.884e-02
age60-69                                 -3.290e-01  7.197e-01  1.580e-02
yod2010-2022                              3.549e-01  1.426e+00  1.279e-02
raceHispanic (All Races)                  9.038e-02  1.095e+00  2.640e-02
raceNon-Hispanic American Indian/Alaska Native  2.347e-02  1.024e+00  1.003e-01
raceNon-Hispanic Asian or Pacific Islander    -3.176e-02  9.687e-01  4.634e-02
raceNon-Hispanic Black                    1.643e-01  1.179e+00  3.889e-02
raceNon-Hispanic Unknown Race            -4.740e-02  9.537e-01  1.493e-01
eventCancer death                         2.207e+01  3.863e+09  1.154e+02
eventOther death                          1.558e-01  1.169e+00  2.373e+02
subsiteTrunk                             -5.414e-03  9.946e-01  1.570e-02
subsiteArm                               -8.475e-02  9.187e-01  1.743e-02
subsiteLeg                               -8.197e-02  9.213e-01  1.820e-02
                                               z Pr(>|z|)
sexMale                                    1.360 0.173743
stageRegional                             39.343  < 2e-16 ***
stageDistant                              75.198  < 2e-16 ***
stageUnstaged                             15.260  < 2e-16 ***
age<50                                   -20.666  < 2e-16 ***
age50-59                                 -21.289  < 2e-16 ***
age60-69                                 -20.817  < 2e-16 ***
yod2010-2022                              27.746  < 2e-16 ***
raceHispanic (All Races)                   3.424 0.000617 ***
raceNon-Hispanic American Indian/Alaska Native   0.234 0.815015
raceNon-Hispanic Asian or Pacific Islander     -0.685 0.493139
raceNon-Hispanic Black                     4.224 2.40e-05 ***
raceNon-Hispanic Unknown Race             -0.317 0.750891
```

```
eventCancer death                              0.191 0.848262    
eventOther death                               0.001 0.999476    
subsiteTrunk                                  -0.345 0.730132    
subsiteArm                                    -4.862 1.16e-06 ***
subsiteLeg                                    -4.504 6.66e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                                             exp(coef) exp(-coef)  lower .95
sexMale                                      1.019e+00  9.818e-01  9.919e-01
stageRegional                                1.748e+00  5.720e-01  1.700e+00
stageDistant                                 3.889e+00  2.571e-01  3.754e+00
stageUnstaged                                1.466e+00  6.819e-01  1.396e+00
age<50                                       6.557e-01  1.525e+00  6.300e-01
age50-59                                     6.696e-01  1.493e+00  6.454e-01
age60-69                                     7.197e-01  1.390e+00  6.977e-01
yod2010-2022                                 1.426e+00  7.013e-01  1.391e+00
raceHispanic (All Races)                     1.095e+00  9.136e-01  1.039e+00
raceNon-Hispanic American Indian/Alaska Native 1.024e+00  9.768e-01  8.410e-01
raceNon-Hispanic Asian or Pacific Islander   9.687e-01  1.032e+00  8.846e-01
raceNon-Hispanic Black                       1.179e+00  8.485e-01  1.092e+00
raceNon-Hispanic Unknown Race                9.537e-01  1.049e+00  7.118e-01
eventCancer death                            3.863e+09  2.588e-10  2.414e-89
eventOther death                             1.169e+00  8.557e-01 1.150e-202
subsiteTrunk                                 9.946e-01  1.005e+00  9.645e-01
subsiteArm                                   9.187e-01  1.088e+00  8.879e-01
subsiteLeg                                   9.213e-01  1.085e+00  8.890e-01
                                              upper .95
sexMale                                       1.046e+00
stageRegional                                 1.798e+00
stageDistant                                  4.029e+00
stageUnstaged                                 1.540e+00
age<50                                        6.825e-01
age50-59                                      6.948e-01
age60-69                                      7.423e-01
yod2010-2022                                  1.462e+00
raceHispanic (All Races)                      1.153e+00
raceNon-Hispanic American Indian/Alaska Native  1.246e+00
raceNon-Hispanic Asian or Pacific Islander    1.061e+00
raceNon-Hispanic Black                        1.272e+00
raceNon-Hispanic Unknown Race                 1.278e+00
eventCancer death                            6.182e+107
eventOther death                             1.187e+202
subsiteTrunk                                  1.026e+00
subsiteArm                                    9.507e-01
subsiteLeg                                    9.548e-01

Concordance= 0.979  (se = 0 )
Likelihood ratio test= 162841  on 18 df,   p=<2e-16
```

```
Wald test            = 7886  on 18 df,   p=<2e-16
Score (logrank) test = 450381  on 18 df,   p=<2e-16
```

```
# Did the hazard ratio for sex increase or decrease?
# What if we included all the covariates in a multivariable model?
```

```
data_sex_female = data
data_sex_female$sex = relevel(data$sex, ref = "Male")

# By sex female
fit.coxph.sex_female <- coxph(cancer.death.surv.5yr ~ sex,
                              data = data_sex_female)
summary(fit.coxph.sex_female)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ sex, data = data_sex_female)

  n= 416918, number of events= 27689 

              coef exp(coef) se(coef)      z Pr(>|z|)    
sexFemale -0.49774   0.60790  0.01306 -38.12   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

          exp(coef) exp(-coef) lower .95 upper .95
sexFemale    0.6079      1.645    0.5925    0.6237

Concordance= 0.557  (se = 0.001 )
Likelihood ratio test= 1538  on 1 df,   p=<2e-16
Wald test            = 1453  on 1 df,   p=<2e-16
Score (logrank) test = 1484  on 1 df,   p=<2e-16
```

```
# Stage is the most important predictor of stage, so other variable are
# vulnerable to confounding. We should adjust for stage.
fit.coxph.sex_female_stage <- coxph(cancer.death.surv.5yr ~ sex + stage,
                                    data = data_sex_female)
summary(fit.coxph.sex_female_stage)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ sex + stage, data = data_sex_female)

  n= 416918, number of events= 27689 

                  coef exp(coef) se(coef)      z Pr(>|z|)    
sexFemale     -0.40096   0.66968  0.01308 -30.66   <2e-16 ***
stageRegional  2.07687   7.97947  0.01394 148.97   <2e-16 ***
stageDistant   3.20209  24.58386  0.01761 181.84   <2e-16 ***
stageUnstaged  0.69945   2.01265  0.02493  28.06   <2e-16 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

              exp(coef) exp(-coef) lower .95 upper .95
sexFemale        0.6697    1.49325    0.6527    0.6871
stageRegional    7.9795    0.12532    7.7644    8.2005
stageDistant    24.5839    0.04068   23.7499   25.4471
stageUnstaged    2.0127    0.49686    1.9167    2.1134

Concordance= 0.748  (se = 0.002 )
Likelihood ratio test= 33986  on 4 df,   p=<2e-16
Wald test            = 44341  on 4 df,   p=<2e-16
Score (logrank) test = 78629  on 4 df,   p=<2e-16
```

```r
fit.coxph.sex_female_full <- coxph(cancer.death.surv.5yr ~ sex + stage + age + yod + race + event
                                   data = data_sex_female)
```

```
Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :
Loglik converged before variable 14 ; coefficient may be infinite.
```

```r
summary(fit.coxph.sex_female_full)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ sex + stage + age + yod +
    race + event + subsite, data = data_sex_female)

  n= 415909, number of events= 27536
   (1009 observations deleted due to missingness)

                                                   coef  exp(coef)   se(coef)
sexFemale                                    -1.835e-02  9.818e-01  1.349e-02
stageRegional                                 5.586e-01  1.748e+00  1.420e-02
stageDistant                                  1.358e+00  3.889e+00  1.806e-02
stageUnstaged                                 3.828e-01  1.466e+00  2.509e-02
age<50                                       -4.221e-01  6.557e-01  2.042e-02
age50-59                                     -4.010e-01  6.696e-01  1.884e-02
age60-69                                     -3.290e-01  7.197e-01  1.580e-02
yod2010-2022                                  3.549e-01  1.426e+00  1.279e-02
raceHispanic (All Races)                      9.038e-02  1.095e+00  2.640e-02
raceNon-Hispanic American Indian/Alaska Native  2.347e-02  1.024e+00  1.003e-01
raceNon-Hispanic Asian or Pacific Islander   -3.176e-02  9.687e-01  4.634e-02
raceNon-Hispanic Black                        1.643e-01  1.179e+00  3.889e-02
raceNon-Hispanic Unknown Race                -4.740e-02  9.537e-01  1.493e-01
eventCancer death                             2.207e+01  3.863e+09  1.154e+02
eventOther death                              1.558e-01  1.169e+00  2.373e+02
subsiteTrunk                                 -5.414e-03  9.946e-01  1.570e-02
subsiteArm                                   -8.475e-02  9.187e-01  1.743e-02
subsiteLeg                                   -8.197e-02  9.213e-01  1.820e-02
                                                   z Pr(>|z|)
```

```
sexFemale                                       -1.360 0.173743    
stageRegional                                   39.343  < 2e-16 ***
stageDistant                                    75.198  < 2e-16 ***
stageUnstaged                                   15.260  < 2e-16 ***
age<50                                         -20.666  < 2e-16 ***
age50-59                                       -21.289  < 2e-16 ***
age60-69                                       -20.817  < 2e-16 ***
yod2010-2022                                    27.746  < 2e-16 ***
raceHispanic (All Races)                         3.424 0.000617 ***
raceNon-Hispanic American Indian/Alaska Native   0.234 0.815015    
raceNon-Hispanic Asian or Pacific Islander      -0.685 0.493139    
raceNon-Hispanic Black                           4.224 2.40e-05 ***
raceNon-Hispanic Unknown Race                   -0.317 0.750891    
eventCancer death                                0.191 0.848266    
eventOther death                                 0.001 0.999476    
subsiteTrunk                                    -0.345 0.730132    
subsiteArm                                      -4.862 1.16e-06 ***
subsiteLeg                                      -4.504 6.66e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                                               exp(coef) exp(-coef)  lower .95
sexFemale                                      9.818e-01  1.019e+00  9.562e-01
stageRegional                                  1.748e+00  5.720e-01  1.700e+00
stageDistant                                   3.889e+00  2.571e-01  3.754e+00
stageUnstaged                                  1.466e+00  6.819e-01  1.396e+00
age<50                                         6.557e-01  1.525e+00  6.300e-01
age50-59                                       6.696e-01  1.493e+00  6.454e-01
age60-69                                       7.197e-01  1.390e+00  6.977e-01
yod2010-2022                                   1.426e+00  7.013e-01  1.391e+00
raceHispanic (All Races)                       1.095e+00  9.136e-01  1.039e+00
raceNon-Hispanic American Indian/Alaska Native 1.024e+00  9.768e-01  8.410e-01
raceNon-Hispanic Asian or Pacific Islander     9.687e-01  1.032e+00  8.846e-01
raceNon-Hispanic Black                         1.179e+00  8.485e-01  1.092e+00
raceNon-Hispanic Unknown Race                  9.537e-01  1.049e+00  7.118e-01
eventCancer death                              3.863e+09  2.588e-10  2.397e-89
eventOther death                               1.169e+00  8.557e-01 1.146e-202
subsiteTrunk                                   9.946e-01  1.005e+00  9.645e-01
subsiteArm                                     9.187e-01  1.088e+00  8.879e-01
subsiteLeg                                     9.213e-01  1.085e+00  8.890e-01
                                                upper .95
sexFemale                                       1.008e+00
stageRegional                                   1.798e+00
stageDistant                                    4.029e+00
stageUnstaged                                   1.540e+00
age<50                                          6.825e-01
age50-59                                        6.948e-01
age60-69                                        7.423e-01
yod2010-2022                                    1.462e+00
raceHispanic (All Races)                        1.153e+00
raceNon-Hispanic American Indian/Alaska Native  1.246e+00
```

```
raceNon-Hispanic Asian or Pacific Islander      1.061e+00
raceNon-Hispanic Black                          1.272e+00
raceNon-Hispanic Unknown Race                   1.278e+00
eventCancer death                              6.225e+107
eventOther death                               1.191e+202
subsiteTrunk                                    1.026e+00
subsiteArm                                      9.507e-01
subsiteLeg                                      9.548e-01

Concordance= 0.979  (se = 0 )
Likelihood ratio test= 162841  on 18 df,   p=<2e-16
Wald test            = 7886  on 18 df,   p=<2e-16
Score (logrank) test = 450381  on 18 df,   p=<2e-16
```

```
# Define a censoring variable at 5 years survival
# By race (black)
fit.coxph.race_black <- coxph(cancer.death.surv.5yr ~ race,
                              data = data)
summary(fit.coxph.race_black)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ race, data = data)

  n= 416918, number of events= 27689

                                                coef exp(coef) se(coef)
raceHispanic (All Races)                     0.41304   1.51141  0.02591
raceNon-Hispanic American Indian/Alaska Native  0.28546   1.33037  0.10020
raceNon-Hispanic Asian or Pacific Islander      0.52299   1.68706  0.04608
raceNon-Hispanic Black                          0.72638   2.06758  0.03833
raceNon-Hispanic Unknown Race                  -3.08919   0.04554  0.14921
                                                    z Pr(>|z|)
raceHispanic (All Races)                        15.942  < 2e-16 ***
raceNon-Hispanic American Indian/Alaska Native   2.849  0.00439 **
raceNon-Hispanic Asian or Pacific Islander      11.349  < 2e-16 ***
raceNon-Hispanic Black                          18.952  < 2e-16 ***
raceNon-Hispanic Unknown Race                  -20.704  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                                               exp(coef) exp(-coef) lower .95
raceHispanic (All Races)                         1.51141     0.6616   1.43657
raceNon-Hispanic American Indian/Alaska Native   1.33037     0.7517   1.09315
raceNon-Hispanic Asian or Pacific Islander       1.68706     0.5927   1.54136
raceNon-Hispanic Black                           2.06758     0.4837   1.91795
raceNon-Hispanic Unknown Race                    0.04554    21.9592   0.03399
                                               upper .95
raceHispanic (All Races)                         1.59014
raceNon-Hispanic American Indian/Alaska Native   1.61906
raceNon-Hispanic Asian or Pacific Islander       1.84653
```

```
raceNon-Hispanic Black                             2.22888
raceNon-Hispanic Unknown Race                      0.06101

Concordance= 0.536  (se = 0.001 )
Likelihood ratio test= 2256  on 5 df,   p=<2e-16
Wald test            = 1147  on 5 df,   p=<2e-16
Score (logrank) test = 1691  on 5 df,   p=<2e-16
```

```r
# The hazard ratio is exp(ceof). Is is significant?

# Stage is the most important predictor of stage, so other variable are
# vulnerable to confounding. We should adjust for stage.
fit.coxph.race_black_stage <- coxph(cancer.death.surv.5yr ~ race + stage,
                                    data = data)
summary(fit.coxph.race_black_stage)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ race + stage, data = data)

  n= 416918, number of events= 27689 

                                               coef exp(coef) se(coef)
raceHispanic (All Races)                    0.11721   1.12435  0.02599
raceNon-Hispanic American Indian/Alaska Native  0.16138   1.17513  0.10021
raceNon-Hispanic Asian or Pacific Islander  0.12071   1.12830  0.04618
raceNon-Hispanic Black                      0.29428   1.34216  0.03847
raceNon-Hispanic Unknown Race              -2.78183   0.06193  0.14936
stageRegional                               2.06089   7.85294  0.01396
stageDistant                                3.18799  24.23956  0.01773
stageUnstaged                               0.77506   2.17073  0.02496
                                                  z Pr(>|z|)    
raceHispanic (All Races)                      4.509 6.51e-06 ***
raceNon-Hispanic American Indian/Alaska Native   1.610  0.10730    
raceNon-Hispanic Asian or Pacific Islander    2.614  0.00896 ** 
raceNon-Hispanic Black                        7.649 2.02e-14 ***
raceNon-Hispanic Unknown Race               -18.625  < 2e-16 ***
stageRegional                               147.594  < 2e-16 ***
stageDistant                                179.808  < 2e-16 ***
stageUnstaged                                31.054  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                                            exp(coef) exp(-coef) lower .95
raceHispanic (All Races)                      1.12435    0.88940   1.06851
raceNon-Hispanic American Indian/Alaska Native   1.17513    0.85097   0.96558
raceNon-Hispanic Asian or Pacific Islander    1.12830    0.88629   1.03065
raceNon-Hispanic Black                        1.34216    0.74507   1.24468
raceNon-Hispanic Unknown Race                 0.06193   16.14853   0.04621
stageRegional                                 7.85294    0.12734   7.64094
stageDistant                                 24.23956    0.04125  23.41171
```

```
stageUnstaged                                   2.17073   0.46068   2.06709
                                              upper .95
raceHispanic (All Races)                        1.18312
raceNon-Hispanic American Indian/Alaska Native  1.43015
raceNon-Hispanic Asian or Pacific Islander      1.23520
raceNon-Hispanic Black                          1.44728
raceNon-Hispanic Unknown Race                   0.08299
stageRegional                                   8.07082
stageDistant                                   25.09669
stageUnstaged                                   2.27955

Concordance= 0.736  (se = 0.002 )
Likelihood ratio test= 34174  on 8 df,   p=<2e-16
Wald test            = 43106  on 8 df,   p=<2e-16
Score (logrank) test = 78274  on 8 df,   p=<2e-16
```

```r
# Stage is the most important predictor of stage, so other variable are
# vulnerable to confounding. We should adjust for stage.
fit.coxph.race_black_full <- coxph(cancer.death.surv.5yr ~ race + stage + age + yod + sex + event
                                   data = data)
```

```
Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :
Loglik converged before variable 14 ; coefficient may be infinite.
```

```r
summary(fit.coxph.race_black_full)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ race + stage + age +
    yod + sex + event + subsite, data = data)

  n= 415909, number of events= 27536
   (1009 observations deleted due to missingness)

                                                     coef  exp(coef)   se(coef)
raceHispanic (All Races)                        9.038e-02  1.095e+00  2.640e-02
raceNon-Hispanic American Indian/Alaska Native  2.347e-02  1.024e+00  1.003e-01
raceNon-Hispanic Asian or Pacific Islander     -3.176e-02  9.687e-01  4.634e-02
raceNon-Hispanic Black                          1.643e-01  1.179e+00  3.889e-02
raceNon-Hispanic Unknown Race                  -4.740e-02  9.537e-01  1.493e-01
stageRegional                                   5.586e-01  1.748e+00  1.420e-02
stageDistant                                    1.358e+00  3.889e+00  1.806e-02
stageUnstaged                                   3.828e-01  1.466e+00  2.509e-02
age<50                                         -4.221e-01  6.557e-01  2.042e-02
age50-59                                       -4.010e-01  6.696e-01  1.884e-02
age60-69                                       -3.290e-01  7.197e-01  1.580e-02
yod2010-2022                                    3.549e-01  1.426e+00  1.279e-02
sexMale                                         1.835e-02  1.019e+00  1.349e-02
eventCancer death                               2.207e+01  3.863e+09  1.154e+02
```

```
eventOther death                               1.558e-01  1.169e+00  2.373e+02
subsiteTrunk                                  -5.414e-03  9.946e-01  1.570e-02
subsiteArm                                    -8.475e-02  9.187e-01  1.743e-02
subsiteLeg                                    -8.197e-02  9.213e-01  1.820e-02
                                                    z Pr(>|z|)    
raceHispanic (All Races)                        3.424 0.000617 ***
raceNon-Hispanic American Indian/Alaska Native   0.234 0.815015    
raceNon-Hispanic Asian or Pacific Islander     -0.685 0.493139    
raceNon-Hispanic Black                          4.224 2.40e-05 ***
raceNon-Hispanic Unknown Race                  -0.317 0.750891    
stageRegional                                  39.343  < 2e-16 ***
stageDistant                                   75.198  < 2e-16 ***
stageUnstaged                                  15.260  < 2e-16 ***
age<50                                        -20.666  < 2e-16 ***
age50-59                                      -21.289  < 2e-16 ***
age60-69                                      -20.817  < 2e-16 ***
yod2010-2022                                   27.746  < 2e-16 ***
sexMale                                         1.360 0.173743    
eventCancer death                               0.191 0.848262    
eventOther death                                0.001 0.999476    
subsiteTrunk                                   -0.345 0.730132    
subsiteArm                                     -4.862 1.16e-06 ***
subsiteLeg                                     -4.504 6.66e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                                              exp(coef) exp(-coef)  lower .95
raceHispanic (All Races)                      1.095e+00  9.136e-01  1.039e+00
raceNon-Hispanic American Indian/Alaska Native 1.024e+00  9.768e-01  8.410e-01
raceNon-Hispanic Asian or Pacific Islander     9.687e-01  1.032e+00  8.846e-01
raceNon-Hispanic Black                        1.179e+00  8.485e-01  1.092e+00
raceNon-Hispanic Unknown Race                 9.537e-01  1.049e+00  7.118e-01
stageRegional                                 1.748e+00  5.720e-01  1.700e+00
stageDistant                                  3.889e+00  2.571e-01  3.754e+00
stageUnstaged                                 1.466e+00  6.819e-01  1.396e+00
age<50                                        6.557e-01  1.525e+00  6.300e-01
age50-59                                      6.696e-01  1.493e+00  6.454e-01
age60-69                                      7.197e-01  1.390e+00  6.977e-01
yod2010-2022                                  1.426e+00  7.013e-01  1.391e+00
sexMale                                       1.019e+00  9.818e-01  9.919e-01
eventCancer death                             3.863e+09  2.588e-10  2.414e-89
eventOther death                              1.169e+00  8.557e-01 1.150e-202
subsiteTrunk                                  9.946e-01  1.005e+00  9.645e-01
subsiteArm                                    9.187e-01  1.088e+00  8.879e-01
subsiteLeg                                    9.213e-01  1.085e+00  8.890e-01
                                               upper .95
raceHispanic (All Races)                       1.153e+00
raceNon-Hispanic American Indian/Alaska Native  1.246e+00
raceNon-Hispanic Asian or Pacific Islander      1.061e+00
raceNon-Hispanic Black                         1.272e+00
raceNon-Hispanic Unknown Race                  1.278e+00
```

```
stageRegional                              1.798e+00
stageDistant                               4.029e+00
stageUnstaged                              1.540e+00
age<50                                     6.825e-01
age50-59                                   6.948e-01
age60-69                                   7.423e-01
yod2010-2022                               1.462e+00
sexMale                                    1.046e+00
eventCancer death                         6.182e+107
eventOther death                          1.187e+202
subsiteTrunk                               1.026e+00
subsiteArm                                 9.507e-01
subsiteLeg                                 9.548e-01

Concordance= 0.979  (se = 0 )
Likelihood ratio test= 162841  on 18 df,   p=<2e-16
Wald test            = 7886  on 18 df,   p=<2e-16
Score (logrank) test = 450381  on 18 df,   p=<2e-16
```

```
# Did the hazard ratio for sex increase or decrease?
# What if we included all the covariates in a multivariable model?
```

```
data_race_white = data
data_race_white$race = relevel(data$race, ref = "Non-Hispanic Black")

# Define a censoring variable at 5 years survival
# By race white
fit.coxph.race_white <- coxph(cancer.death.surv.5yr ~ race,
                              data = data_race_white)
summary(fit.coxph.race_white)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ race, data = data_race_white)

  n= 416918, number of events= 27689

                                                coef exp(coef) se(coef)
raceNon-Hispanic White                      -0.72638   0.48366  0.03833
raceHispanic (All Races)                    -0.31334   0.73100  0.04538
raceNon-Hispanic American Indian/Alaska Native -0.44092   0.64344  0.10690
raceNon-Hispanic Asian or Pacific Islander  -0.20339   0.81596  0.05926
raceNon-Hispanic Unknown Race               -3.81556   0.02203  0.15379
                                                  z Pr(>|z|)
raceNon-Hispanic White                      -18.952  < 2e-16 ***
raceHispanic (All Races)                     -6.905 5.04e-12 ***
raceNon-Hispanic American Indian/Alaska Native  -4.124 3.72e-05 ***
raceNon-Hispanic Asian or Pacific Islander   -3.432 0.000599 ***
raceNon-Hispanic Unknown Race               -24.811  < 2e-16 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                                              exp(coef) exp(-coef) lower .95
raceNon-Hispanic White                          0.48366      2.068   0.44866
raceHispanic (All Races)                        0.73100      1.368   0.66879
raceNon-Hispanic American Indian/Alaska Native  0.64344      1.554   0.52181
raceNon-Hispanic Asian or Pacific Islander      0.81596      1.226   0.72648
raceNon-Hispanic Unknown Race                   0.02203     45.402   0.01629
                                              upper .95
raceNon-Hispanic White                          0.52139
raceHispanic (All Races)                        0.79900
raceNon-Hispanic American Indian/Alaska Native  0.79343
raceNon-Hispanic Asian or Pacific Islander      0.91646
raceNon-Hispanic Unknown Race                   0.02977

Concordance= 0.536  (se = 0.001 )
Likelihood ratio test= 2256  on 5 df,   p=<2e-16
Wald test            = 1147  on 5 df,   p=<2e-16
Score (logrank) test = 1691  on 5 df,   p=<2e-16
```

```r
# The hazard ratio is exp(ceof). Is is significant?

# Stage is the most important predictor of stage, so other variable are
# vulnerable to confounding. We should adjust for stage.
fit.coxph.race_white_stage <- coxph(cancer.death.surv.5yr ~ race + stage,
                                    data = data_race_white)
summary(fit.coxph.race_white_stage)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ race + stage, data = data_race_white)

  n= 416918, number of events= 27689

                                                   coef exp(coef) se(coef)
raceNon-Hispanic White                         -0.29428   0.74507  0.03847
raceHispanic (All Races)                       -0.17707   0.83772  0.04541
raceNon-Hispanic American Indian/Alaska Native -0.13290   0.87555  0.10694
raceNon-Hispanic Asian or Pacific Islander     -0.17357   0.84066  0.05926
raceNon-Hispanic Unknown Race                  -3.07611   0.04614  0.15399
stageRegional                                   2.06089   7.85294  0.01396
stageDistant                                    3.18799  24.23956  0.01773
stageUnstaged                                   0.77506   2.17073  0.02496
                                                     z Pr(>|z|)
raceNon-Hispanic White                          -7.649 2.02e-14 ***
raceHispanic (All Races)                        -3.900 9.63e-05 ***
raceNon-Hispanic American Indian/Alaska Native  -1.243   0.2140
raceNon-Hispanic Asian or Pacific Islander      -2.929   0.0034 **
raceNon-Hispanic Unknown Race                  -19.976  < 2e-16 ***
stageRegional                                  147.594  < 2e-16 ***
stageDistant                                   179.808  < 2e-16 ***
```

```
stageUnstaged                                 31.054  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                                              exp(coef) exp(-coef) lower .95
raceNon-Hispanic White                          0.74507    1.34216   0.69095
raceHispanic (All Races)                        0.83772    1.19372   0.76639
raceNon-Hispanic American Indian/Alaska Native  0.87555    1.14214   0.70999
raceNon-Hispanic Asian or Pacific Islander      0.84066    1.18954   0.74847
raceNon-Hispanic Unknown Race                   0.04614   21.67395   0.03412
stageRegional                                   7.85294    0.12734   7.64094
stageDistant                                   24.23956    0.04125  23.41171
stageUnstaged                                   2.17073    0.46068   2.06709
                                              upper .95
raceNon-Hispanic White                          0.80342
raceHispanic (All Races)                        0.91569
raceNon-Hispanic American Indian/Alaska Native  1.07972
raceNon-Hispanic Asian or Pacific Islander      0.94420
raceNon-Hispanic Unknown Race                   0.06239
stageRegional                                   8.07082
stageDistant                                   25.09669
stageUnstaged                                   2.27955

Concordance= 0.736  (se = 0.002 )
Likelihood ratio test= 34174  on 8 df,   p=<2e-16
Wald test            = 43106  on 8 df,   p=<2e-16
Score (logrank) test = 78274  on 8 df,   p=<2e-16
```

```r
# Stage is the most important predictor of stage, so other variable are
# vulnerable to confounding. We should adjust for stage.
fit.coxph.race_white_full <- coxph(cancer.death.surv.5yr ~ race + stage + age + yod + sex + event
                                   data = data_race_white)
```

```
Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :
Loglik converged before variable 14 ; coefficient may be infinite.
```

```r
summary(fit.coxph.race_white_full)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ race + stage + age +
    yod + sex + event + subsite, data = data_race_white)

  n= 415909, number of events= 27536
   (1009 observations deleted due to missingness)

                                                    coef  exp(coef)   se(coef)
raceNon-Hispanic White                        -1.643e-01  8.485e-01  3.889e-02
raceHispanic (All Races)                      -7.388e-02  9.288e-01  4.555e-02
```

```
raceNon-Hispanic American Indian/Alaska Native -1.408e-01  8.687e-01  1.070e-01
raceNon-Hispanic Asian or Pacific Islander     -1.960e-01  8.220e-01  5.940e-02
raceNon-Hispanic Unknown Race                  -2.117e-01  8.092e-01  1.540e-01
stageRegional                                   5.586e-01  1.748e+00  1.420e-02
stageDistant                                    1.358e+00  3.889e+00  1.806e-02
stageUnstaged                                   3.828e-01  1.466e+00  2.509e-02
age<50                                         -4.221e-01  6.557e-01  2.042e-02
age50-59                                       -4.010e-01  6.696e-01  1.884e-02
age60-69                                       -3.290e-01  7.197e-01  1.580e-02
yod2010-2022                                    3.549e-01  1.426e+00  1.279e-02
sexMale                                         1.835e-02  1.019e+00  1.349e-02
eventCancer death                               2.207e+01  3.863e+09  1.154e+02
eventOther death                                1.558e-01  1.169e+00  2.373e+02
subsiteTrunk                                   -5.414e-03  9.946e-01  1.570e-02
subsiteArm                                     -8.475e-02  9.187e-01  1.743e-02
subsiteLeg                                     -8.197e-02  9.213e-01  1.820e-02
                                                     z Pr(>|z|)
raceNon-Hispanic White                          -4.224 2.40e-05 ***
raceHispanic (All Races)                        -1.622 0.104804
raceNon-Hispanic American Indian/Alaska Native  -1.315 0.188443
raceNon-Hispanic Asian or Pacific Islander      -3.300 0.000968 ***
raceNon-Hispanic Unknown Race                   -1.374 0.169346
stageRegional                                   39.343  < 2e-16 ***
stageDistant                                    75.198  < 2e-16 ***
stageUnstaged                                   15.260  < 2e-16 ***
age<50                                         -20.666  < 2e-16 ***
age50-59                                       -21.289  < 2e-16 ***
age60-69                                       -20.817  < 2e-16 ***
yod2010-2022                                    27.746  < 2e-16 ***
sexMale                                          1.360 0.173743
eventCancer death                                0.191 0.848273
eventOther death                                 0.001 0.999476
subsiteTrunk                                    -0.345 0.730132
subsiteArm                                      -4.862 1.16e-06 ***
subsiteLeg                                      -4.504 6.66e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                                               exp(coef) exp(-coef)  lower .95
raceNon-Hispanic White                         8.485e-01  1.179e+00  7.863e-01
raceHispanic (All Races)                       9.288e-01  1.077e+00  8.495e-01
raceNon-Hispanic American Indian/Alaska Native 8.687e-01  1.151e+00  7.043e-01
raceNon-Hispanic Asian or Pacific Islander     8.220e-01  1.217e+00  7.317e-01
raceNon-Hispanic Unknown Race                  8.092e-01  1.236e+00  5.984e-01
stageRegional                                  1.748e+00  5.720e-01  1.700e+00
stageDistant                                   3.889e+00  2.571e-01  3.754e+00
stageUnstaged                                  1.466e+00  6.819e-01  1.396e+00
age<50                                         6.557e-01  1.525e+00  6.300e-01
age50-59                                       6.696e-01  1.493e+00  6.454e-01
age60-69                                       7.197e-01  1.390e+00  6.977e-01
yod2010-2022                                   1.426e+00  7.013e-01  1.391e+00
```

```
sexMale                                        1.019e+00  9.818e-01  9.919e-01
eventCancer death                              3.863e+09  2.588e-10  2.375e-89
eventOther death                               1.169e+00  8.557e-01 1.141e-202
subsiteTrunk                                   9.946e-01  1.005e+00  9.645e-01
subsiteArm                                     9.187e-01  1.088e+00  8.879e-01
subsiteLeg                                     9.213e-01  1.085e+00  8.890e-01
                                                upper .95
raceNon-Hispanic White                          9.157e-01
raceHispanic (All Races)                        1.016e+00
raceNon-Hispanic American Indian/Alaska Native  1.071e+00
raceNon-Hispanic Asian or Pacific Islander      9.235e-01
raceNon-Hispanic Unknown Race                   1.094e+00
stageRegional                                   1.798e+00
stageDistant                                    4.029e+00
stageUnstaged                                   1.540e+00
age<50                                          6.825e-01
age50-59                                        6.948e-01
age60-69                                        7.423e-01
yod2010-2022                                    1.462e+00
sexMale                                         1.046e+00
eventCancer death                              6.285e+107
eventOther death                               1.197e+202
subsiteTrunk                                    1.026e+00
subsiteArm                                      9.507e-01
subsiteLeg                                      9.548e-01

Concordance= 0.979  (se = 0 )
Likelihood ratio test= 162841  on 18 df,   p=<2e-16
Wald test            = 7886  on 18 df,   p=<2e-16
Score (logrank) test = 450381  on 18 df,   p=<2e-16
```

```r
# Did the hazard ratio for sex increase or decrease?
# What if we included all the covariates in a multivariable model?
```

```r
# Define a censoring variable at 5 years survival
# By subsite
fit.coxph.subsite <- coxph(cancer.death.surv.5yr ~ subsite,
                           data = data)
summary(fit.coxph.subsite)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ subsite, data = data)

  n= 415909, number of events= 27536
   (1009 observations deleted due to missingness)

                 coef exp(coef) se(coef)      z Pr(>|z|)
subsiteTrunk -0.48210   0.61749  0.01535 -31.40   <2e-16 ***
subsiteArm   -0.65129   0.52137  0.01733 -37.58   <2e-16 ***
```

```
subsiteLeg   -0.29678   0.74321  0.01738 -17.08   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

             exp(coef) exp(-coef) lower .95 upper .95
subsiteTrunk    0.6175      1.619    0.5992    0.6364
subsiteArm      0.5214      1.918    0.5040    0.5394
subsiteLeg      0.7432      1.346    0.7183    0.7690

Concordance= 0.567  (se = 0.002 )
Likelihood ratio test= 1680  on 3 df,   p=<2e-16
Wald test            = 1717  on 3 df,   p=<2e-16
Score (logrank) test = 1758  on 3 df,   p=<2e-16
```

```
fit.coxph.subsite_stage <- coxph(cancer.death.surv.5yr ~ subsite + stage,
                                 data = data)
summary(fit.coxph.subsite_stage)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ subsite + stage, data = data)

  n= 415909, number of events= 27536
   (1009 observations deleted due to missingness)

                  coef exp(coef) se(coef)      z Pr(>|z|)
subsiteTrunk  -0.29545   0.74419  0.01542 -19.16   <2e-16 ***
subsiteArm    -0.41841   0.65809  0.01744 -24.00   <2e-16 ***
subsiteLeg    -0.24136   0.78556  0.01741 -13.87   <2e-16 ***
stageRegional  2.07430   7.95895  0.01402 147.94   <2e-16 ***
stageDistant   3.18260  24.10938  0.01775 179.26   <2e-16 ***
stageUnstaged  0.68300   1.97981  0.02504  27.28   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

              exp(coef) exp(-coef) lower .95 upper .95
subsiteTrunk     0.7442    1.34373    0.7220    0.7670
subsiteArm       0.6581    1.51955    0.6360    0.6810
subsiteLeg       0.7856    1.27298    0.7592    0.8128
stageRegional    7.9590    0.12564    7.7432    8.1807
stageDistant    24.1094    0.04148   23.2849   24.9631
stageUnstaged    1.9798    0.50510    1.8850    2.0794

Concordance= 0.752  (se = 0.002 )
Likelihood ratio test= 33562  on 6 df,   p=<2e-16
Wald test            = 44062  on 6 df,   p=<2e-16
Score (logrank) test = 78345  on 6 df,   p=<2e-16
```

```
fit.coxph.subsite_full <- coxph(cancer.death.surv.5yr ~ subsite + stage + age + yod + sex + event
```

```
                                      data = data)
```

```
Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :
Loglik converged before variable 12 ; coefficient may be infinite.
```

```
summary(fit.coxph.subsite_full)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ subsite + stage + age +
    yod + sex + event + race, data = data)

  n= 415909, number of events= 27536
   (1009 observations deleted due to missingness)

                                                    coef  exp(coef)   se(coef)
subsiteTrunk                                  -5.414e-03  9.946e-01  1.570e-02
subsiteArm                                    -8.475e-02  9.187e-01  1.743e-02
subsiteLeg                                    -8.197e-02  9.213e-01  1.820e-02
stageRegional                                  5.586e-01  1.748e+00  1.420e-02
stageDistant                                   1.358e+00  3.889e+00  1.806e-02
stageUnstaged                                  3.828e-01  1.466e+00  2.509e-02
age<50                                        -4.221e-01  6.557e-01  2.042e-02
age50-59                                      -4.010e-01  6.696e-01  1.884e-02
age60-69                                      -3.290e-01  7.197e-01  1.580e-02
yod2010-2022                                   3.549e-01  1.426e+00  1.279e-02
sexMale                                        1.835e-02  1.019e+00  1.349e-02
eventCancer death                              2.207e+01  3.863e+09  1.154e+02
eventOther death                               1.558e-01  1.169e+00  2.373e+02
raceHispanic (All Races)                       9.038e-02  1.095e+00  2.640e-02
raceNon-Hispanic American Indian/Alaska Native  2.347e-02  1.024e+00  1.003e-01
raceNon-Hispanic Asian or Pacific Islander    -3.176e-02  9.687e-01  4.634e-02
raceNon-Hispanic Black                         1.643e-01  1.179e+00  3.889e-02
raceNon-Hispanic Unknown Race                 -4.740e-02  9.537e-01  1.493e-01
                                                    z Pr(>|z|)
subsiteTrunk                                   -0.345 0.730132
subsiteArm                                     -4.862 1.16e-06 ***
subsiteLeg                                     -4.504 6.66e-06 ***
stageRegional                                  39.343  < 2e-16 ***
stageDistant                                   75.198  < 2e-16 ***
stageUnstaged                                  15.260  < 2e-16 ***
age<50                                        -20.666  < 2e-16 ***
age50-59                                      -21.289  < 2e-16 ***
age60-69                                      -20.817  < 2e-16 ***
yod2010-2022                                   27.746  < 2e-16 ***
sexMale                                         1.360 0.173743
eventCancer death                               0.191 0.848262
eventOther death                                0.001 0.999476
raceHispanic (All Races)                        3.424 0.000617 ***
```

```
raceNon-Hispanic American Indian/Alaska Native   0.234 0.815015
raceNon-Hispanic Asian or Pacific Islander      -0.685 0.493139
raceNon-Hispanic Black                           4.224 2.40e-05 ***
raceNon-Hispanic Unknown Race                   -0.317 0.750891
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                                               exp(coef) exp(-coef)  lower .95
subsiteTrunk                                   9.946e-01  1.005e+00  9.645e-01
subsiteArm                                     9.187e-01  1.088e+00  8.879e-01
subsiteLeg                                     9.213e-01  1.085e+00  8.890e-01
stageRegional                                  1.748e+00  5.720e-01  1.700e+00
stageDistant                                   3.889e+00  2.571e-01  3.754e+00
stageUnstaged                                  1.466e+00  6.819e-01  1.396e+00
age<50                                         6.557e-01  1.525e+00  6.300e-01
age50-59                                       6.696e-01  1.493e+00  6.454e-01
age60-69                                       7.197e-01  1.390e+00  6.977e-01
yod2010-2022                                   1.426e+00  7.013e-01  1.391e+00
sexMale                                        1.019e+00  9.818e-01  9.919e-01
eventCancer death                              3.863e+09  2.588e-10  2.414e-89
eventOther death                               1.169e+00  8.557e-01 1.150e-202
raceHispanic (All Races)                       1.095e+00  9.136e-01  1.039e+00
raceNon-Hispanic American Indian/Alaska Native 1.024e+00  9.768e-01  8.410e-01
raceNon-Hispanic Asian or Pacific Islander     9.687e-01  1.032e+00  8.846e-01
raceNon-Hispanic Black                         1.179e+00  8.485e-01  1.092e+00
raceNon-Hispanic Unknown Race                  9.537e-01  1.049e+00  7.118e-01
                                                upper .95
subsiteTrunk                                    1.026e+00
subsiteArm                                      9.507e-01
subsiteLeg                                      9.548e-01
stageRegional                                   1.798e+00
stageDistant                                    4.029e+00
stageUnstaged                                   1.540e+00
age<50                                          6.825e-01
age50-59                                        6.948e-01
age60-69                                        7.423e-01
yod2010-2022                                    1.462e+00
sexMale                                         1.046e+00
eventCancer death                              6.182e+107
eventOther death                               1.187e+202
raceHispanic (All Races)                        1.153e+00
raceNon-Hispanic American Indian/Alaska Native  1.246e+00
raceNon-Hispanic Asian or Pacific Islander      1.061e+00
raceNon-Hispanic Black                          1.272e+00
raceNon-Hispanic Unknown Race                   1.278e+00

Concordance= 0.979  (se = 0 )
Likelihood ratio test= 162841  on 18 df,   p=<2e-16
Wald test            = 7886  on 18 df,   p=<2e-16
Score (logrank) test = 450381  on 18 df,   p=<2e-16
```

```
# Define a censoring variable at 5 years survival
# By age
fit.coxph.age <- coxph(cancer.death.surv.5yr ~ age,
                       data = data)
summary(fit.coxph.age)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ age, data = data)

  n= 416918, number of events= 27689 

             coef exp(coef) se(coef)      z Pr(>|z|)    
age<50   -1.18514   0.30570  0.01981 -59.81   <2e-16 ***
age50-59 -0.87930   0.41507  0.01848 -47.58   <2e-16 ***
age60-69 -0.65710   0.51835  0.01561 -42.09   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

         exp(coef) exp(-coef) lower .95 upper .95
age<50      0.3057      3.271    0.2941    0.3178
age50-59    0.4151      2.409    0.4003    0.4304
age60-69    0.5184      1.929    0.5027    0.5345

Concordance= 0.624  (se = 0.002 )
Likelihood ratio test= 5820  on 3 df,   p=<2e-16
Wald test            = 5520  on 3 df,   p=<2e-16
Score (logrank) test = 5962  on 3 df,   p=<2e-16
```

```
fit.coxph.age_stage <- coxph(cancer.death.surv.5yr ~ age + stage,
                             data = data)
summary(fit.coxph.age_stage)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ age + stage, data = data)

  n= 416918, number of events= 27689 

                  coef exp(coef) se(coef)      z Pr(>|z|)    
age<50        -1.11101   0.32923  0.01985 -55.98   <2e-16 ***
age50-59      -0.83230   0.43505  0.01850 -45.00   <2e-16 ***
age60-69      -0.61754   0.53927  0.01562 -39.54   <2e-16 ***
stageRegional  2.08241   8.02381  0.01394 149.44   <2e-16 ***
stageDistant   3.17271  23.87204  0.01761 180.19   <2e-16 ***
stageUnstaged  0.64147   1.89928  0.02494  25.72   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

              exp(coef) exp(-coef) lower .95 upper .95
age<50           0.3292    3.03742    0.3167    0.3423
```

```
age50-59          0.4350     2.29860     0.4196     0.4511
age60-69          0.5393     1.85437     0.5230     0.5560
stageRegional     8.0238     0.12463     7.8076     8.2460
stageDistant     23.8720     0.04189    23.0623    24.7102
stageUnstaged     1.8993     0.52652     1.8087     1.9944

Concordance= 0.791  (se = 0.001 )
Likelihood ratio test= 38114  on 6 df,   p=<2e-16
Wald test            = 48172  on 6 df,   p=<2e-16
Score (logrank) test = 82915  on 6 df,   p=<2e-16
```

```
fit.coxph.age_full <- coxph(cancer.death.surv.5yr ~ age + stage + subsite + yod + sex + event + ra
                            data = data)
```

```
Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :
Loglik converged before variable 12 ; coefficient may be infinite.
```

```
summary(fit.coxph.age_full)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ age + stage + subsite +
    yod + sex + event + race, data = data)

  n= 415909, number of events= 27536
   (1009 observations deleted due to missingness)

                                                  coef  exp(coef)   se(coef)
age<50                                      -4.221e-01  6.557e-01  2.042e-02
age50-59                                    -4.010e-01  6.696e-01  1.884e-02
age60-69                                    -3.290e-01  7.197e-01  1.580e-02
stageRegional                                5.586e-01  1.748e+00  1.420e-02
stageDistant                                 1.358e+00  3.889e+00  1.806e-02
stageUnstaged                                3.828e-01  1.466e+00  2.509e-02
subsiteTrunk                                -5.414e-03  9.946e-01  1.570e-02
subsiteArm                                  -8.475e-02  9.187e-01  1.743e-02
subsiteLeg                                  -8.197e-02  9.213e-01  1.820e-02
yod2010-2022                                 3.549e-01  1.426e+00  1.279e-02
sexMale                                      1.835e-02  1.019e+00  1.349e-02
eventCancer death                            2.207e+01  3.863e+09  1.154e+02
eventOther death                             1.558e-01  1.169e+00  2.373e+02
raceHispanic (All Races)                     9.038e-02  1.095e+00  2.640e-02
raceNon-Hispanic American Indian/Alaska Native  2.347e-02  1.024e+00  1.003e-01
raceNon-Hispanic Asian or Pacific Islander  -3.176e-02  9.687e-01  4.634e-02
raceNon-Hispanic Black                       1.643e-01  1.179e+00  3.889e-02
raceNon-Hispanic Unknown Race               -4.740e-02  9.537e-01  1.493e-01
                                                  z Pr(>|z|)
age<50                                      -20.666  < 2e-16 ***
age50-59                                    -21.289  < 2e-16 ***
```

```
age60-69                                      -20.817  < 2e-16 ***
stageRegional                                  39.343  < 2e-16 ***
stageDistant                                   75.198  < 2e-16 ***
stageUnstaged                                  15.260  < 2e-16 ***
subsiteTrunk                                   -0.345 0.730132    
subsiteArm                                     -4.862 1.16e-06 ***
subsiteLeg                                     -4.504 6.66e-06 ***
yod2010-2022                                   27.746  < 2e-16 ***
sexMale                                         1.360 0.173743    
eventCancer death                               0.191 0.848262    
eventOther death                                0.001 0.999476    
raceHispanic (All Races)                        3.424 0.000617 ***
raceNon-Hispanic American Indian/Alaska Native  0.234 0.815015    
raceNon-Hispanic Asian or Pacific Islander     -0.685 0.493139    
raceNon-Hispanic Black                          4.224 2.40e-05 ***
raceNon-Hispanic Unknown Race                  -0.317 0.750891    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                                               exp(coef) exp(-coef)  lower .95
age<50                                         6.557e-01  1.525e+00  6.300e-01
age50-59                                       6.696e-01  1.493e+00  6.454e-01
age60-69                                       7.197e-01  1.390e+00  6.977e-01
stageRegional                                  1.748e+00  5.720e-01  1.700e+00
stageDistant                                   3.889e+00  2.571e-01  3.754e+00
stageUnstaged                                  1.466e+00  6.819e-01  1.396e+00
subsiteTrunk                                   9.946e-01  1.005e+00  9.645e-01
subsiteArm                                     9.187e-01  1.088e+00  8.879e-01
subsiteLeg                                     9.213e-01  1.085e+00  8.890e-01
yod2010-2022                                   1.426e+00  7.013e-01  1.391e+00
sexMale                                        1.019e+00  9.818e-01  9.919e-01
eventCancer death                              3.863e+09  2.588e-10  2.414e-89
eventOther death                               1.169e+00  8.557e-01 1.150e-202
raceHispanic (All Races)                       1.095e+00  9.136e-01  1.039e+00
raceNon-Hispanic American Indian/Alaska Native 1.024e+00  9.768e-01  8.410e-01
raceNon-Hispanic Asian or Pacific Islander     9.687e-01  1.032e+00  8.846e-01
raceNon-Hispanic Black                         1.179e+00  8.485e-01  1.092e+00
raceNon-Hispanic Unknown Race                  9.537e-01  1.049e+00  7.118e-01
                                                upper .95
age<50                                          6.825e-01
age50-59                                        6.948e-01
age60-69                                        7.423e-01
stageRegional                                   1.798e+00
stageDistant                                    4.029e+00
stageUnstaged                                   1.540e+00
subsiteTrunk                                    1.026e+00
subsiteArm                                      9.507e-01
subsiteLeg                                      9.548e-01
yod2010-2022                                    1.462e+00
sexMale                                         1.046e+00
eventCancer death                              6.182e+107
```

```
eventOther death                              1.187e+202
raceHispanic (All Races)                       1.153e+00
raceNon-Hispanic American Indian/Alaska Native  1.246e+00
raceNon-Hispanic Asian or Pacific Islander     1.061e+00
raceNon-Hispanic Black                         1.272e+00
raceNon-Hispanic Unknown Race                  1.278e+00

Concordance= 0.979  (se = 0 )
Likelihood ratio test= 162841  on 18 df,   p=<2e-16
Wald test            = 7886  on 18 df,   p=<2e-16
Score (logrank) test = 450381  on 18 df,   p=<2e-16
```

```
data_age_70 <- data
data_age_70$age <- relevel(data$age, ref = "<50")

# Define a censoring variable at 5 years survival
# By age greater 70
fit.coxph.age_70 <- coxph(cancer.death.surv.5yr ~ age,
                          data = data_age_70)
summary(fit.coxph.age_70)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ age, data = data_age_70)

  n= 416918, number of events= 27689

            coef exp(coef) se(coef)     z Pr(>|z|)
age70+   1.18514   3.27113  0.01981 59.81   <2e-16 ***
age50-59 0.30583   1.35775  0.02455 12.46   <2e-16 ***
age60-69 0.52804   1.69560  0.02248 23.49   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

         exp(coef) exp(-coef) lower .95 upper .95
age70+       3.271     0.3057     3.147     3.401
age50-59     1.358     0.7365     1.294     1.425
age60-69     1.696     0.5898     1.623     1.772

Concordance= 0.624  (se = 0.002 )
Likelihood ratio test= 5820  on 3 df,   p=<2e-16
Wald test            = 5520  on 3 df,   p=<2e-16
Score (logrank) test = 5962  on 3 df,   p=<2e-16
```

```
fit.coxph.age_70_stage <- coxph(cancer.death.surv.5yr ~ age + stage,
                                data = data_age_70)
summary(fit.coxph.age_70_stage)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ age + stage, data = data_age_70)
```

```
  n= 416918, number of events= 27689

                   coef exp(coef) se(coef)      z Pr(>|z|)
age70+          1.11101   3.03742  0.01985  55.98   <2e-16 ***
age50-59        0.27871   1.32142  0.02456  11.35   <2e-16 ***
age60-69        0.49347   1.63798  0.02249  21.94   <2e-16 ***
stageRegional   2.08241   8.02381  0.01394 149.44   <2e-16 ***
stageDistant    3.17271  23.87204  0.01761 180.19   <2e-16 ***
stageUnstaged   0.64147   1.89928  0.02494  25.72   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

              exp(coef) exp(-coef) lower .95 upper .95
age70+            3.037    0.32923     2.922     3.158
age50-59          1.321    0.75676     1.259     1.387
age60-69          1.638    0.61051     1.567     1.712
stageRegional     8.024    0.12463     7.808     8.246
stageDistant     23.872    0.04189    23.062    24.710
stageUnstaged     1.899    0.52652     1.809     1.994

Concordance= 0.791  (se = 0.001 )
Likelihood ratio test= 38114  on 6 df,   p=<2e-16
Wald test            = 48172  on 6 df,   p=<2e-16
Score (logrank) test = 82915  on 6 df,   p=<2e-16
```

```
fit.coxph.age_70_full <- coxph(cancer.death.surv.5yr ~ age + stage + subsite + yod + sex + event -
                               data = data_age_70)
```

```
Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :
Loglik converged before variable 12 ; coefficient may be infinite.
```

```
summary(fit.coxph.age_70_full)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ age + stage + subsite +
    yod + sex + event + race, data = data_age_70)

  n= 415909, number of events= 27536
   (1009 observations deleted due to missingness)

                                                  coef  exp(coef)   se(coef)
age70+                                       4.221e-01  1.525e+00  2.042e-02
age50-59                                     2.108e-02  1.021e+00  2.469e-02
age60-69                                     9.309e-02  1.098e+00  2.277e-02
stageRegional                                5.586e-01  1.748e+00  1.420e-02
stageDistant                                 1.358e+00  3.889e+00  1.806e-02
stageUnstaged                                3.828e-01  1.466e+00  2.509e-02
```

```
subsiteTrunk                                 -5.414e-03  9.946e-01  1.570e-02
subsiteArm                                   -8.475e-02  9.187e-01  1.743e-02
subsiteLeg                                   -8.197e-02  9.213e-01  1.820e-02
yod2010-2022                                  3.549e-01  1.426e+00  1.279e-02
sexMale                                       1.835e-02  1.019e+00  1.349e-02
eventCancer death                             2.207e+01  3.863e+09  1.154e+02
eventOther death                              1.558e-01  1.169e+00  2.373e+02
raceHispanic (All Races)                      9.038e-02  1.095e+00  2.640e-02
raceNon-Hispanic American Indian/Alaska Native  2.347e-02  1.024e+00  1.003e-01
raceNon-Hispanic Asian or Pacific Islander   -3.176e-02  9.687e-01  4.634e-02
raceNon-Hispanic Black                        1.643e-01  1.179e+00  3.889e-02
raceNon-Hispanic Unknown Race                -4.740e-02  9.537e-01  1.493e-01
                                                   z Pr(>|z|)
age70+                                       20.666  < 2e-16 ***
age50-59                                      0.854 0.393293
age60-69                                      4.088 4.35e-05 ***
stageRegional                                39.343  < 2e-16 ***
stageDistant                                 75.198  < 2e-16 ***
stageUnstaged                                15.260  < 2e-16 ***
subsiteTrunk                                 -0.345 0.730132
subsiteArm                                   -4.862 1.16e-06 ***
subsiteLeg                                   -4.504 6.66e-06 ***
yod2010-2022                                 27.746  < 2e-16 ***
sexMale                                       1.360 0.173743
eventCancer death                             0.191 0.848272
eventOther death                              0.001 0.999476
raceHispanic (All Races)                      3.424 0.000617 ***
raceNon-Hispanic American Indian/Alaska Native  0.234 0.815015
raceNon-Hispanic Asian or Pacific Islander   -0.685 0.493139
raceNon-Hispanic Black                        4.224 2.40e-05 ***
raceNon-Hispanic Unknown Race                -0.317 0.750891
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                                             exp(coef) exp(-coef)  lower .95
age70+                                       1.525e+00  6.557e-01  1.465e+00
age50-59                                     1.021e+00  9.791e-01  9.731e-01
age60-69                                     1.098e+00  9.111e-01  1.050e+00
stageRegional                                1.748e+00  5.720e-01  1.700e+00
stageDistant                                 3.889e+00  2.571e-01  3.754e+00
stageUnstaged                                1.466e+00  6.819e-01  1.396e+00
subsiteTrunk                                 9.946e-01  1.005e+00  9.645e-01
subsiteArm                                   9.187e-01  1.088e+00  8.879e-01
subsiteLeg                                   9.213e-01  1.085e+00  8.890e-01
yod2010-2022                                 1.426e+00  7.013e-01  1.391e+00
sexMale                                      1.019e+00  9.818e-01  9.919e-01
eventCancer death                            3.863e+09  2.588e-10  2.377e-89
eventOther death                             1.169e+00  8.557e-01 1.142e-202
raceHispanic (All Races)                     1.095e+00  9.136e-01  1.039e+00
raceNon-Hispanic American Indian/Alaska Native 1.024e+00  9.768e-01  8.410e-01
raceNon-Hispanic Asian or Pacific Islander   9.687e-01  1.032e+00  8.846e-01
```

```
raceNon-Hispanic Black                       1.179e+00  8.485e-01  1.092e+00
raceNon-Hispanic Unknown Race                9.537e-01  1.049e+00  7.118e-01
                                              upper .95
age70+                                        1.587e+00
age50-59                                      1.072e+00
age60-69                                      1.148e+00
stageRegional                                 1.798e+00
stageDistant                                  4.029e+00
stageUnstaged                                 1.540e+00
subsiteTrunk                                  1.026e+00
subsiteArm                                    9.507e-01
subsiteLeg                                    9.548e-01
yod2010-2022                                  1.462e+00
sexMale                                       1.046e+00
eventCancer death                            6.278e+107
eventOther death                             1.196e+202
raceHispanic (All Races)                      1.153e+00
raceNon-Hispanic American Indian/Alaska Native  1.246e+00
raceNon-Hispanic Asian or Pacific Islander    1.061e+00
raceNon-Hispanic Black                        1.272e+00
raceNon-Hispanic Unknown Race                 1.278e+00

Concordance= 0.979  (se = 0 )
Likelihood ratio test= 162841  on 18 df,   p=<2e-16
Wald test            = 7886  on 18 df,   p=<2e-16
Score (logrank) test = 450381  on 18 df,   p=<2e-16
```

```r
data_subsite_hn <- data
data_subsite_hn$subsite <- relevel(data$subsite, ref = "Trunk")

# Define a censoring variable at 5 years survival
# By race white
fit.coxph.subsite_hn <- coxph(cancer.death.surv.5yr ~ subsite,
                              data = data_subsite_hn)
summary(fit.coxph.subsite_hn)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ subsite, data = data_subsite_hn)

  n= 415909, number of events= 27536
   (1009 observations deleted due to missingness)

                         coef exp(coef) se(coef)      z Pr(>|z|)
subsiteHead and Neck  0.48210   1.61947  0.01535 31.398   <2e-16 ***
subsiteArm           -0.16919   0.84435  0.01774 -9.534   <2e-16 ***
subsiteLeg            0.18532   1.20360  0.01779 10.417   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                     exp(coef) exp(-coef) lower .95 upper .95
```

```
subsiteHead and Neck    1.6195     0.6175    1.5715    1.6689
subsiteArm              0.8444     1.1843    0.8155    0.8742
subsiteLeg              1.2036     0.8308    1.1624    1.2463

Concordance= 0.567  (se = 0.002 )
Likelihood ratio test= 1680  on 3 df,   p=<2e-16
Wald test            = 1717  on 3 df,   p=<2e-16
Score (logrank) test = 1758  on 3 df,   p=<2e-16
```

```
fit.coxph.subsite_hn_stage <- coxph(cancer.death.surv.5yr ~ subsite + stage,
                                    data = data_subsite_hn)
summary(fit.coxph.subsite_hn_stage)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ subsite + stage, data = data_subsite_hn)

  n= 415909, number of events= 27536
   (1009 observations deleted due to missingness)

                         coef exp(coef) se(coef)       z Pr(>|z|)
subsiteHead and Neck  0.29545   1.34373  0.01542  19.155  < 2e-16 ***
subsiteArm           -0.12296   0.88430  0.01775  -6.927  4.3e-12 ***
subsiteLeg            0.05410   1.05559  0.01783   3.035  0.00241 **
stageRegional         2.07430   7.95895  0.01402 147.939  < 2e-16 ***
stageDistant          3.18260  24.10938  0.01775 179.264  < 2e-16 ***
stageUnstaged         0.68300   1.97981  0.02504  27.282  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                     exp(coef) exp(-coef) lower .95 upper .95
subsiteHead and Neck    1.3437    0.74419    1.3037    1.3850
subsiteArm              0.8843    1.13084    0.8541    0.9156
subsiteLeg              1.0556    0.94734    1.0193    1.0931
stageRegional           7.9590    0.12564    7.7432    8.1807
stageDistant           24.1094    0.04148   23.2849   24.9631
stageUnstaged           1.9798    0.50510    1.8850    2.0794

Concordance= 0.752  (se = 0.002 )
Likelihood ratio test= 33562  on 6 df,   p=<2e-16
Wald test            = 44062  on 6 df,   p=<2e-16
Score (logrank) test = 78345  on 6 df,   p=<2e-16
```

```
fit.coxph.subsite_hn_full <- coxph(cancer.death.surv.5yr ~ subsite + stage + age + yod + sex + ev
                                   data = data_subsite_hn)
```

```
Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :
Loglik converged before variable 12 ; coefficient may be infinite.
```

```r
summary(fit.coxph.subsite_hn_full)
```

```
Call:
coxph(formula = cancer.death.surv.5yr ~ subsite + stage + age +
    yod + sex + event + race, data = data_subsite_hn)

  n= 415909, number of events= 27536
   (1009 observations deleted due to missingness)

                                                   coef  exp(coef)   se(coef)
subsiteHead and Neck                          5.414e-03  1.005e+00  1.570e-02
subsiteArm                                   -7.933e-02  9.237e-01  1.788e-02
subsiteLeg                                   -7.656e-02  9.263e-01  1.848e-02
stageRegional                                 5.586e-01  1.748e+00  1.420e-02
stageDistant                                  1.358e+00  3.889e+00  1.806e-02
stageUnstaged                                 3.828e-01  1.466e+00  2.509e-02
age<50                                       -4.221e-01  6.557e-01  2.042e-02
age50-59                                     -4.010e-01  6.696e-01  1.884e-02
age60-69                                     -3.290e-01  7.197e-01  1.580e-02
yod2010-2022                                  3.549e-01  1.426e+00  1.279e-02
sexMale                                       1.835e-02  1.019e+00  1.349e-02
eventCancer death                             2.207e+01  3.863e+09  1.154e+02
eventOther death                              1.558e-01  1.169e+00  2.373e+02
raceHispanic (All Races)                      9.038e-02  1.095e+00  2.640e-02
raceNon-Hispanic American Indian/Alaska Native  2.347e-02  1.024e+00  1.003e-01
raceNon-Hispanic Asian or Pacific Islander   -3.176e-02  9.687e-01  4.634e-02
raceNon-Hispanic Black                        1.643e-01  1.179e+00  3.889e-02
raceNon-Hispanic Unknown Race                -4.740e-02  9.537e-01  1.493e-01
                                                   z Pr(>|z|)
subsiteHead and Neck                            0.345 0.730132
subsiteArm                                     -4.437 9.14e-06 ***
subsiteLeg                                     -4.142 3.44e-05 ***
stageRegional                                  39.343  < 2e-16 ***
stageDistant                                   75.198  < 2e-16 ***
stageUnstaged                                  15.260  < 2e-16 ***
age<50                                        -20.666  < 2e-16 ***
age50-59                                      -21.289  < 2e-16 ***
age60-69                                      -20.817  < 2e-16 ***
yod2010-2022                                   27.746  < 2e-16 ***
sexMale                                         1.360 0.173743
eventCancer death                               0.191 0.848271
eventOther death                                0.001 0.999476
raceHispanic (All Races)                        3.424 0.000617 ***
raceNon-Hispanic American Indian/Alaska Native  0.234 0.815015
raceNon-Hispanic Asian or Pacific Islander     -0.685 0.493139
raceNon-Hispanic Black                          4.224 2.40e-05 ***
raceNon-Hispanic Unknown Race                  -0.317 0.750891
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
                                               exp(coef) exp(-coef)  lower .95
subsiteHead and Neck                           1.005e+00  9.946e-01  9.750e-01
subsiteArm                                     9.237e-01  1.083e+00  8.919e-01
subsiteLeg                                     9.263e-01  1.080e+00  8.933e-01
stageRegional                                  1.748e+00  5.720e-01  1.700e+00
stageDistant                                   3.889e+00  2.571e-01  3.754e+00
stageUnstaged                                  1.466e+00  6.819e-01  1.396e+00
age<50                                         6.557e-01  1.525e+00  6.300e-01
age50-59                                       6.696e-01  1.493e+00  6.454e-01
age60-69                                       7.197e-01  1.390e+00  6.977e-01
yod2010-2022                                   1.426e+00  7.013e-01  1.391e+00
sexMale                                        1.019e+00  9.818e-01  9.919e-01
eventCancer death                              3.863e+09  2.588e-10  2.382e-89
eventOther death                               1.169e+00  8.557e-01 1.143e-202
raceHispanic (All Races)                       1.095e+00  9.136e-01  1.039e+00
raceNon-Hispanic American Indian/Alaska Native 1.024e+00  9.768e-01  8.410e-01
raceNon-Hispanic Asian or Pacific Islander     9.687e-01  1.032e+00  8.846e-01
raceNon-Hispanic Black                         1.179e+00  8.485e-01  1.092e+00
raceNon-Hispanic Unknown Race                  9.537e-01  1.049e+00  7.118e-01
                                                upper .95
subsiteHead and Neck                            1.037e+00
subsiteArm                                      9.567e-01
subsiteLeg                                      9.605e-01
stageRegional                                   1.798e+00
stageDistant                                    4.029e+00
stageUnstaged                                   1.540e+00
age<50                                          6.825e-01
age50-59                                        6.948e-01
age60-69                                        7.423e-01
yod2010-2022                                    1.462e+00
sexMale                                         1.046e+00
eventCancer death                              6.266e+107
eventOther death                               1.195e+202
raceHispanic (All Races)                        1.153e+00
raceNon-Hispanic American Indian/Alaska Native  1.246e+00
raceNon-Hispanic Asian or Pacific Islander      1.061e+00
raceNon-Hispanic Black                          1.272e+00
raceNon-Hispanic Unknown Race                   1.278e+00

Concordance= 0.979  (se = 0 )
Likelihood ratio test= 162841  on 18 df,   p=<2e-16
Wald test            = 7886  on 18 df,   p=<2e-16
Score (logrank) test = 450381  on 18 df,   p=<2e-16
```

Censoring the data after 5 years, we compute the Cox proportional hazards models for each covariate in univariable, bivariable (with stage), and multivariable (all covariates) models.

Group	Univariable	Bivariable (with stage)	Multivariable
Age <50 years	0.31 (95%CI: 0.29, 0.32); p<0.001	0.33 (95%CI: 0.32, 0.34); p<0.001	0.66 (95%CI: 0.63, 0.68); p<0.001
Age 50-59 years	0.42 (95%CI: 0.40, 0.43); p<0.001	0.44 (95%CI: 0.42, 0.45); p<0.001	0.67 (95%CI: 0.65, 0.69); p<0.001
Age 60-69 years	0.52 (95%CI: 0.50, 0.53); p<0.001	0.54 (95%CI: 0.52, 0.56); p<0.001	0.72 (95%CI: 0.70, 0.74); p<0.001
Age 70+ years	3.27 (95%CI: 3.15, 3.40); p<0.001	3.04 (95%CI: 2.92, 3.16); p<0.001	1.53 (95%CI: 1.45, 1.59); p<0.001
Men	1.65 (95%CI: 1.60, 1.69); p<0.001	1.49 (95%CI: 1.46, 1.53); p<0.001	0.98 (95%CI: 0.96, 1.01); p = 0.173743
Women	0.61 (95%CI: 0.59, 0.62); p<0.001	0.67 (95%CI: 0.65, 0.69); p<0.001	1.49 (95%CI: 1.46, 1.53); p<0.001
Black	2.07 (95%CI: 1.92, 2.23); p<0.001	1.34 (95%CI: 1.24, 1.45); p<0.001	1.18 (95%CI: 1.09, 1.27); p<0.001
White	0.48 (95%CI: 0.45, 0.53); p<0.001	0.75 (95%CI: 0.69, 0.80); p<0.001	0.85 (95%CI: 0.79, 0.92); p<0.001
Head and Neck	1.62 (95%CI: 1.57, 1.67); p<0.001	1.34 (95%CI: 1.30, 1.39); p<0.001	1.00 (95%CI: 0.98, 1.04); p = 0.730132
Trunk	0.62 (95%CI: 0.60, 0.64); p<0.001	0.74 (95%CI: 0.72, 0.77); p<0.001	0.99 (95%CI: 0.96, 1.03); p = 0.730132
Arm	0.52 (95%CI: 0.50, 0.54); p<0.001	0.66 (95%CI: 0.64, 0.68); p<0.001	0.92 (95%CI: 0.89, 0.95); p<0.001
Leg	0.74 (95%CI: 0.72, 0.77); p<0.001	0.79 (95%CI: 0.76, 0.81); p<0.001	0.92 (95%CI: 0.79, 0.95); p<0.001